Introduction to Linux environment

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Programa Formación Continua, ISCIII

Overview

1. Linux OS
2. Linux file system
3. Basic commands
4. Command line syntax
5. Linux users and privileges

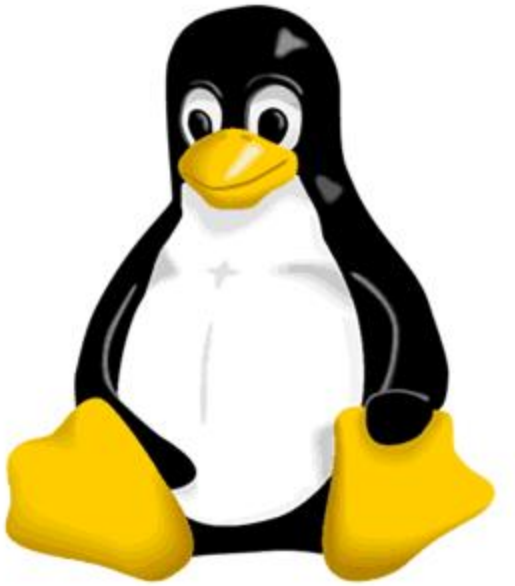
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1. Linux Operating System (OS)

Features:

1. Operating System
2. Open source
3. Linux distributions
4. Multitasking
5. Multi-user

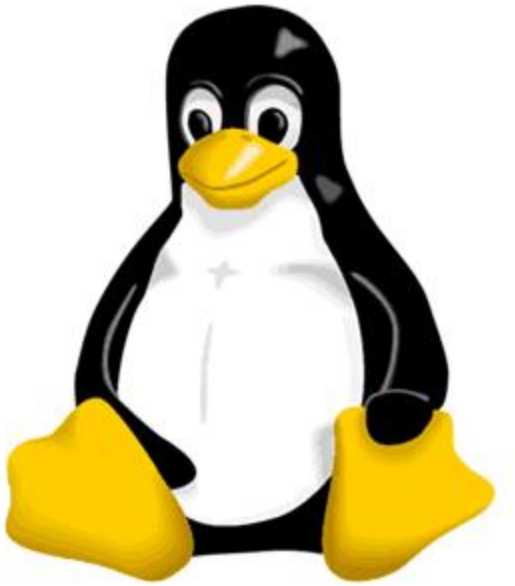


Linux

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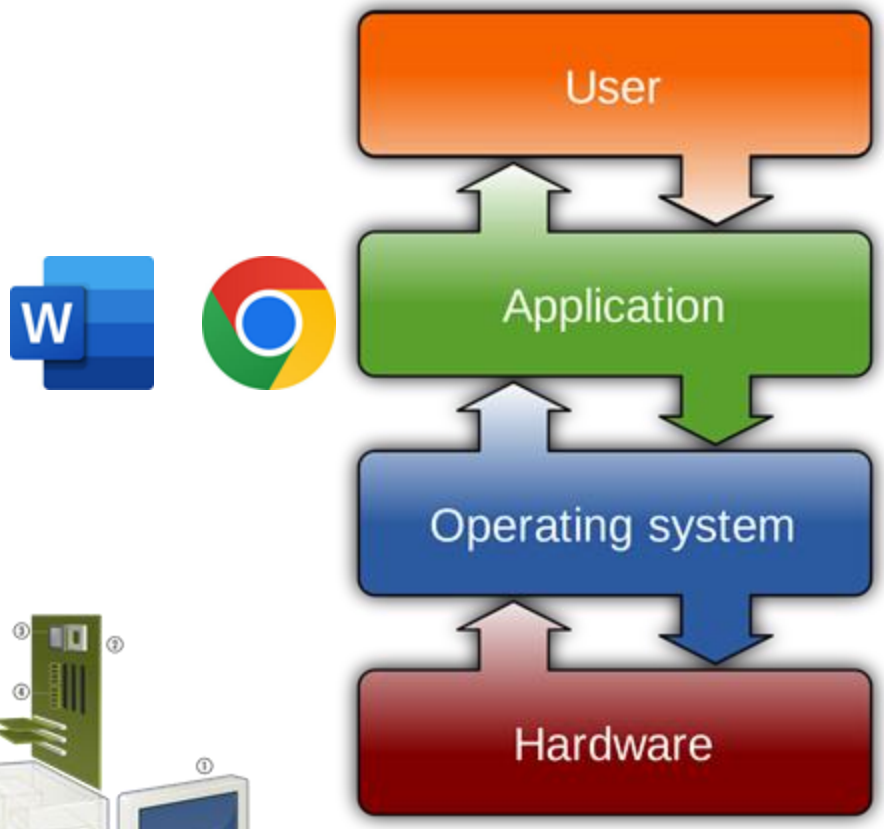


Linux

1.1. Operating System

Operating System (OS):

- **Software** that manages computer hardware and software resources and provides common services for computer programs



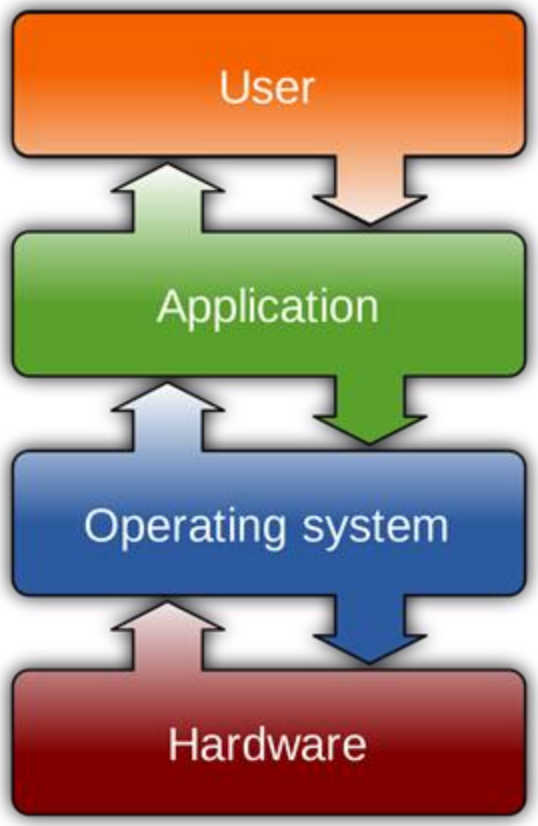
1.1. Operating System

Operating System (OS):

- Software that manages

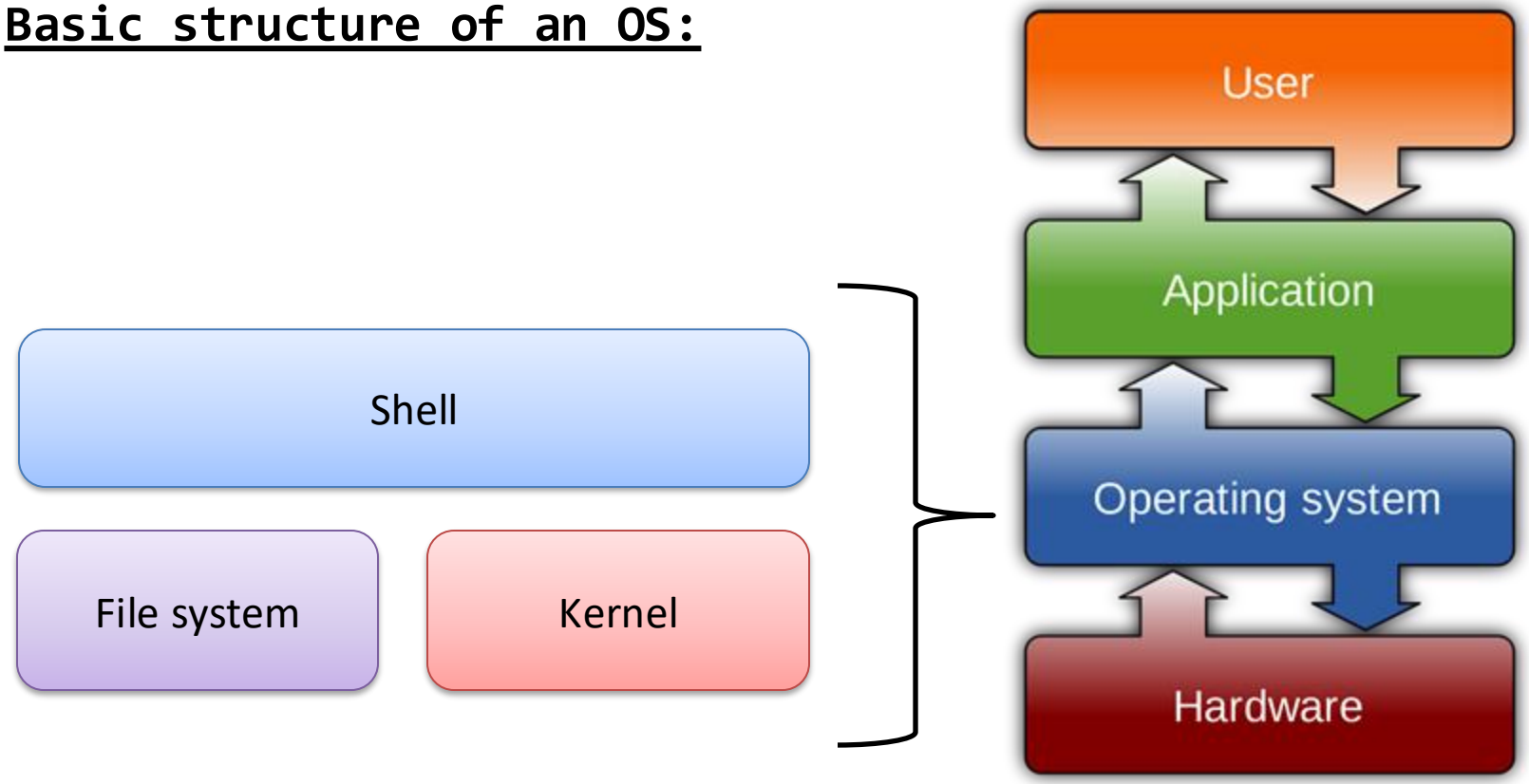
Functions:

- Program execution and control (launch, manage, and supervises running programs)
- Peripheral management (controls devices like keyboard, mouse, screen, printer, USB…)
- User and permission management (creates users, assigns roles, and controls access to files and system functions)
- Error and security management (detects system or hardware failures and protects against unauthorized access)



1.1. Operating System

Basic structure of an OS:



1.1. Operating System

Kernel

It is the foundational layer of an OS. It functions at a basic level, communicating with hardware and managing resources.

1.1. Operating System

File system

It controls how data is stored, manipulated and retrieved. (Part 2)

Kernel

It is the foundational layer of an operating system (OS). It functions at a basic level, communicating with hardware and managing resources.

1.1. Operating System

Shell

Program that provides the traditional, text-only user interface. (Part 3)

File system

It controls how data is stored, manipulated and retrieved. (Part 2)

Kernel

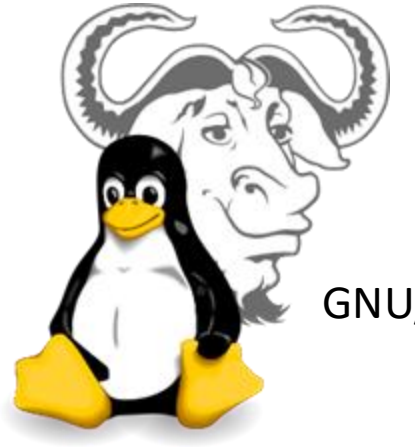
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1.1. Operating System

Operating System (OS):

Windows

Mac OS



GNU/Linux

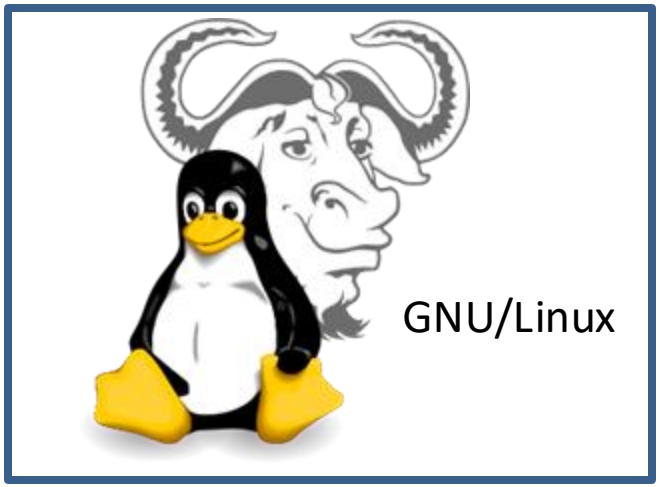
1.1. Operating System

GNU/Linux:

GNU = **G**NU's **N**ot **U**nix project
(Richard Stallman, 1983)
- Shell
- Compiler
- Libraries
- Other utils



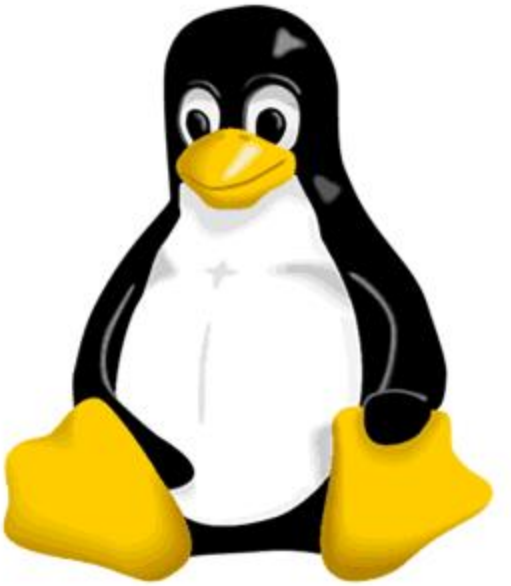
Kernel (Linus Torvalds, 1991)



1. Linux Operating System (OS)

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Linux

2. Open source

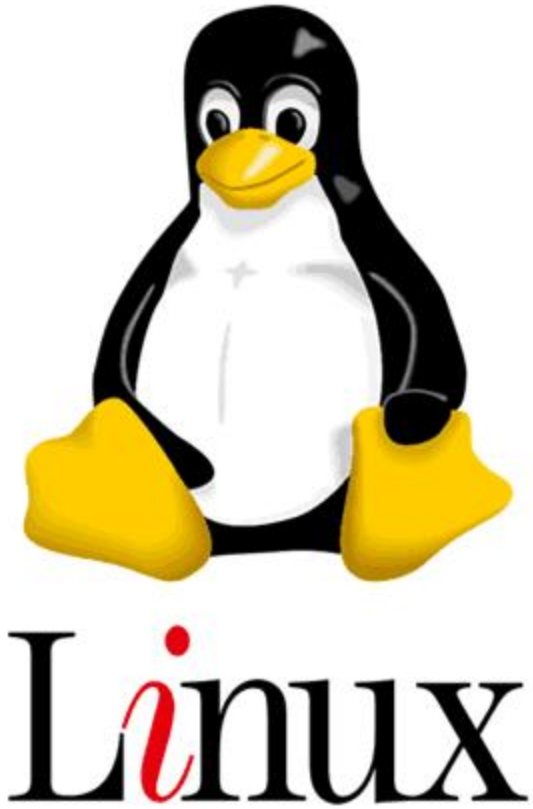
The distribution terms of open-source software must comply with the following criteria:

1. Free Redistribution
2. Source Code
3. Derived Works
4. Integrity of the Author's Source Code
5. No Discrimination Against Persons or Groups
6. No Discrimination Against Fields of Endeavour
7. Distribution of license
8. License Must Not Be Specific to a Product
9. License Must not Restrict Other Software
10. License Must Be Technology-Neutral

1. Linux Operating System (OS)

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3. Linux Distributions

Linux Distributions

- A distro is a Linux kernel based operating system made from a software collection and sometimes a package management system.
- There are distros for a wide variety of platforms.
- A typical Linux distro comprises a Linux kernel, GNU tools and libraries, additional software, documentation, a window system, a window manager, and a desktop environment.
- Most of the included software is free and open-source software made available both as compiled binaries and in source code form, allowing modifications to the original software. This means that you can see how the software works, and even modify it, which is especially useful in science, where reproducibility is key.

3. Linux Distributions

“Distro” - Version Kernel + programs:

- Ubuntu: Beginner-friendly.
- Debian: Known for its stability.
- Fedora: Sponsored by Red Hat, cutting-edge technology.
- Arch Linux: Designed for advanced users, full control over system configuration.
- **CentOS** / Rocky Linux / AlmaLinux: Oriented towards server environments.

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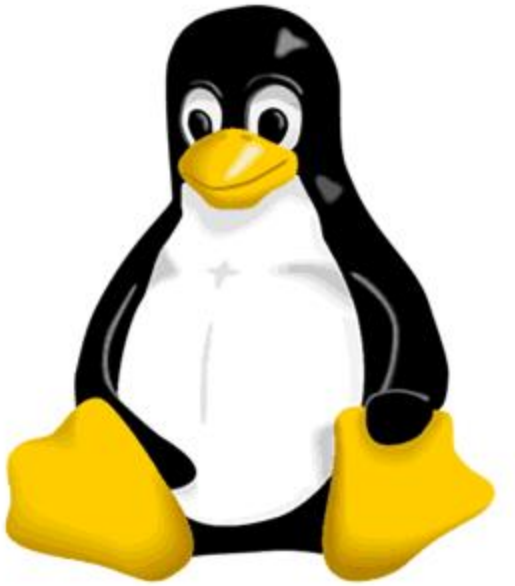


CentOS Stream release 8

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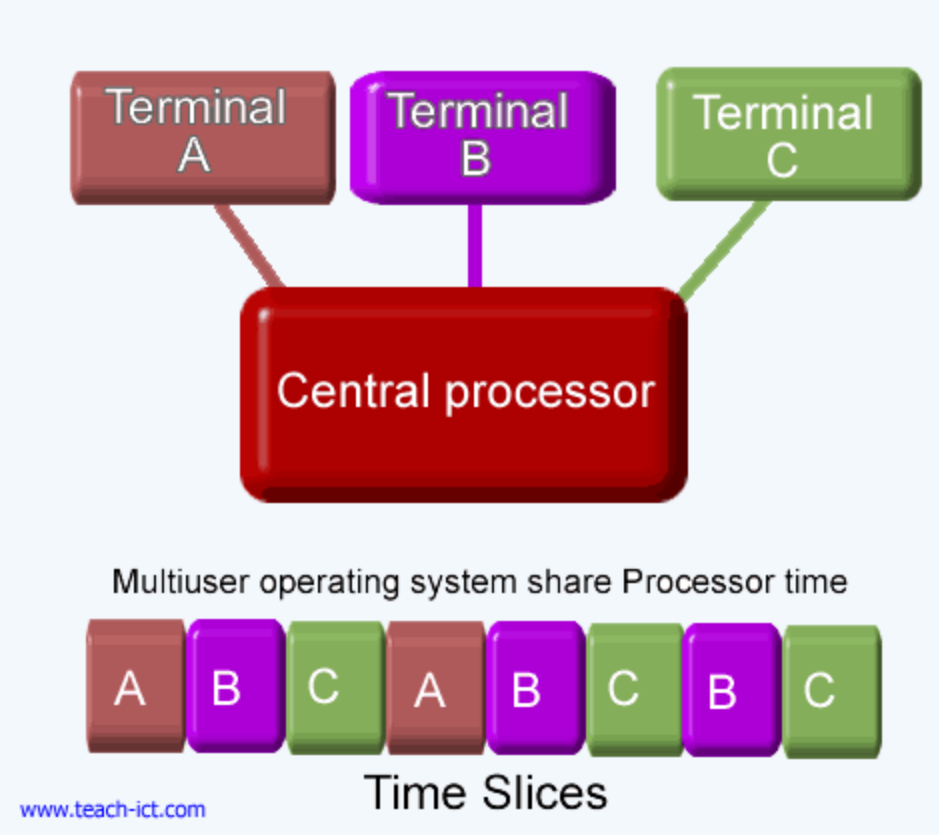


Linux

4. Multi-task

A multitasking operating systems allows a user to perform more than one computer task (such as the operation of an application program) at a time.

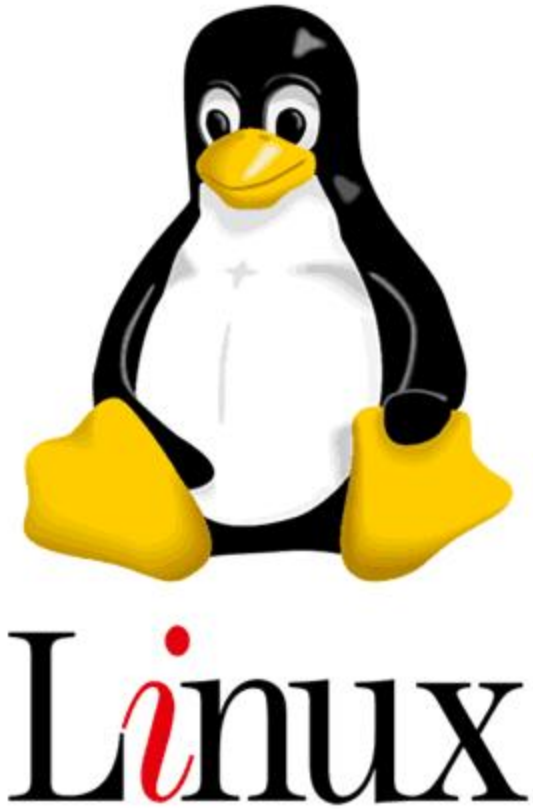
The operating system is able to keep track of where you are in these tasks and go from one to the other without losing information.



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5. Multi-user

- A multi-user operating system is software that allows access by multiple users of a computer.
- An example is a server where multiple remote users have access (such as via a serial port or Secure Shell) to the Unix shell prompt at the same time.

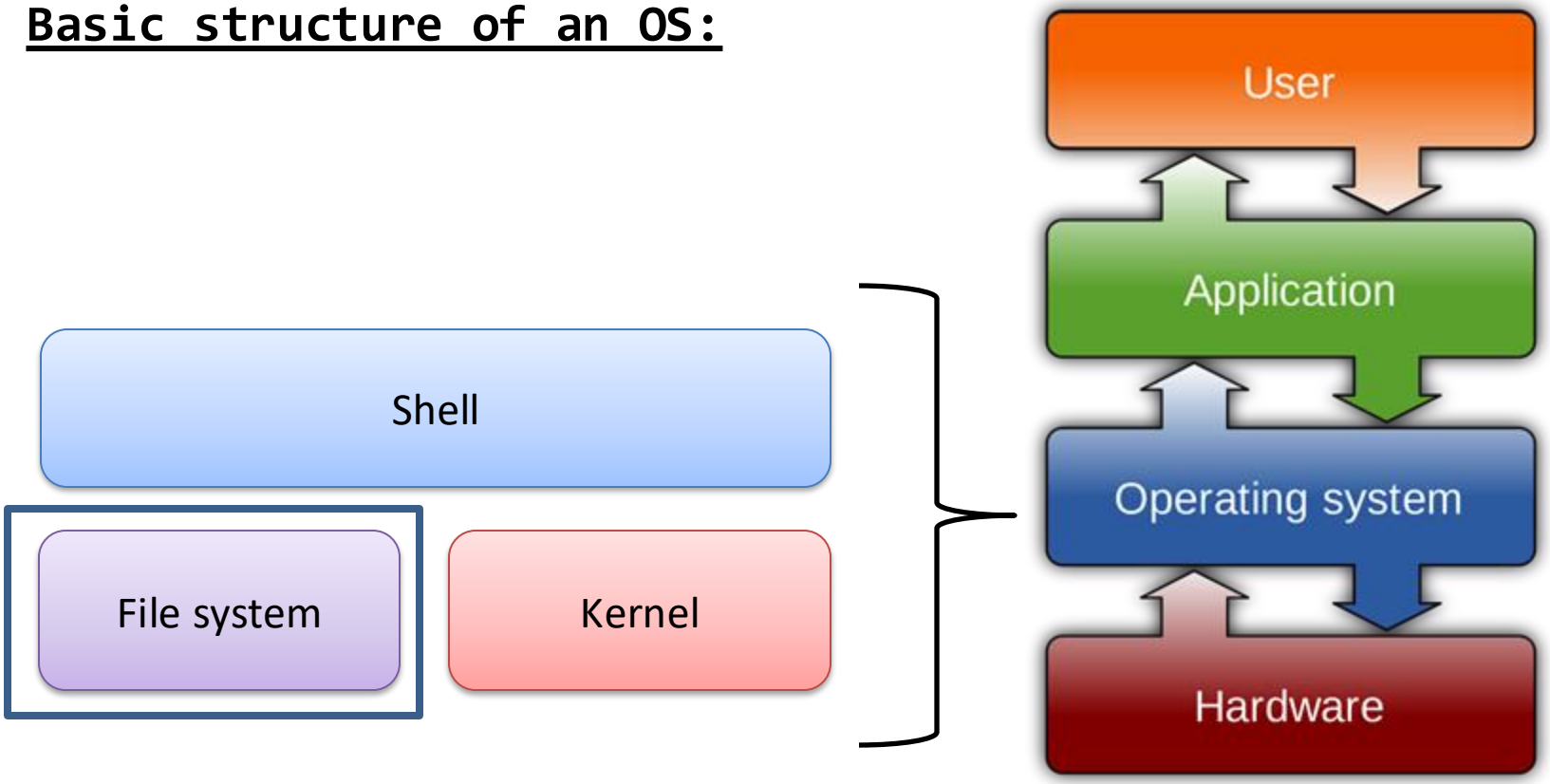


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1.1. Operating System

Basic structure of an OS:



2. Linux File System

1. Features
2. Structure
3. Comparison with Windows
4. Paths

2.1. Linux File System Features

The file system is what gives structure to the data. It **allows**:

- Storing files and folders in an organized way.
- Accessing them by name, path, and permissions.
- Controlling who can read, modify, or execute each file.

When you **save a file** the file system:

- Divides it into physical blocks stored on the disk.
- Associates metadata (name, date, permissions, size, owner).
- Places it in a folder hierarchy to make it easier to locate.

It also **manages**:

- Permissions and ownership
- Access/modification times
- Free and used disk space
- Integrity and error recovery

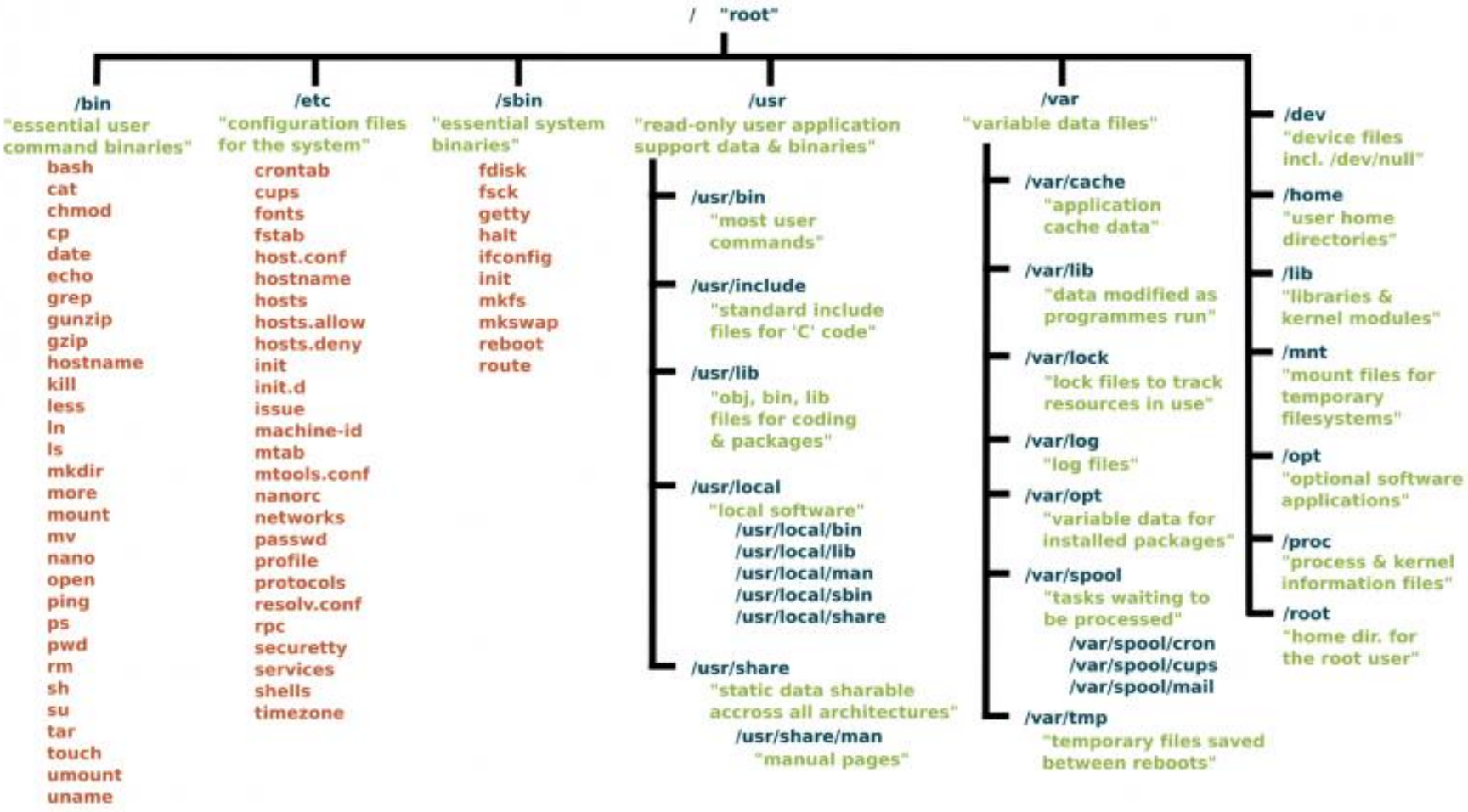
2.1. Linux File System Features

- Root directory (/)
- Everything is a File (devices, directories, and processes)
- Specifying Paths
- Case-Sensitive
- File Extensions and Hidden Files

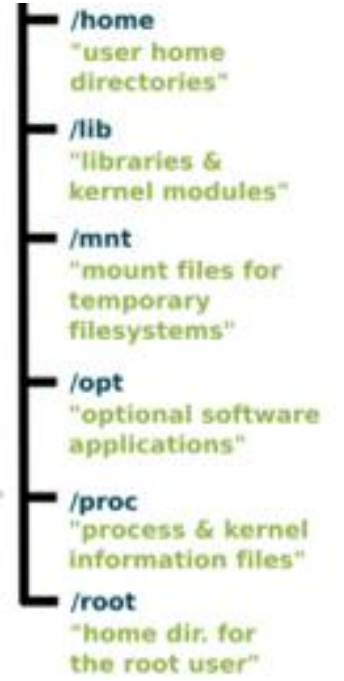
2. Linux File System

1. Features
2. Structure
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2.2. Linux File System Structure



2.2. Linux File System Structure



User data directory

Or /media, directory where external disks are mounted.

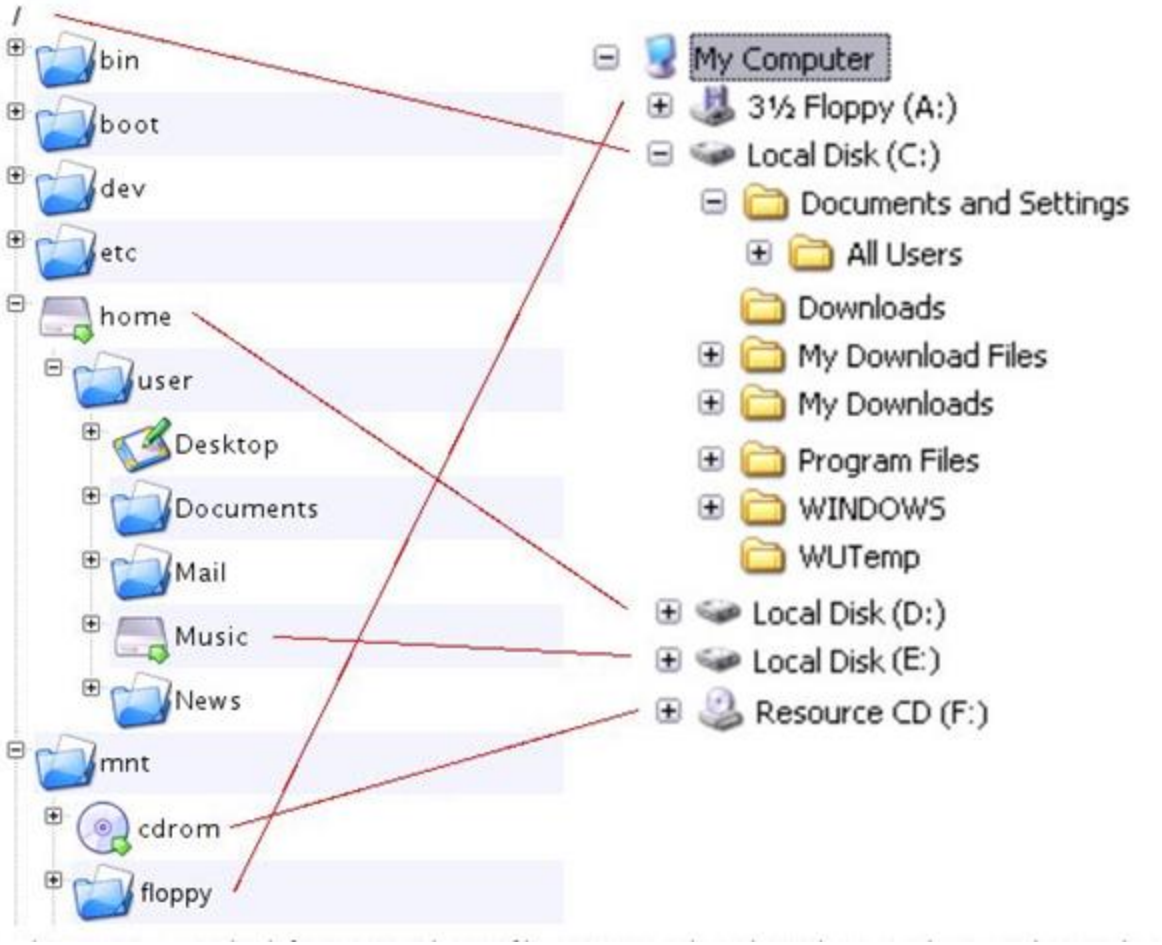
Directory containing additional software. In our case is where bioinformatic software is installed.

2. Linux File System

1. Features
2. Structure
3. Comparison with Windows
4. Paths

2.3. Linux File System Comparison

- Everything “hangs” from root
- Files are classified by type / role instead of unit location
- Files locations in disks are invisible for users



2. Linux File System

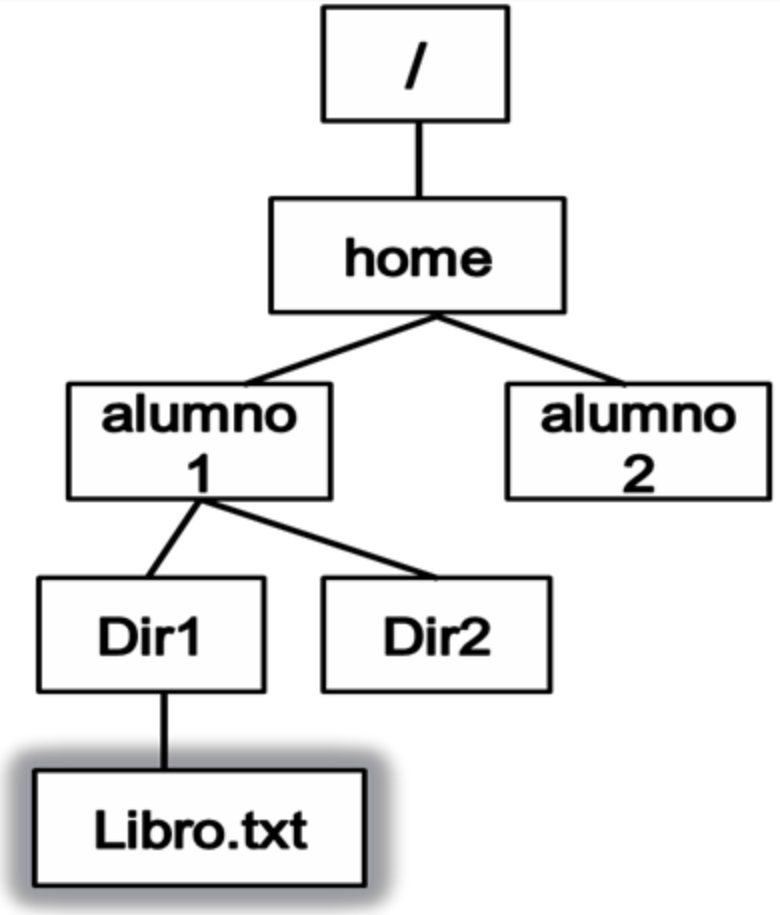
1. Features
2. Structure
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2.4. Linux File System Paths

- PATH:
  - **Absolute path:**
    - Location of a file or directory from the root directory (/).
    - Static.
    - Ej: /home/alumno1/dir1/book.txt
  - **Relative path:**
    - Path related to the present working directory (pwd).
    - Variable
    - Actual pwd = “.”
    - Parent directory = “..”
    - Ej:Path to book.txt
      - From /home: ./alumno1/dir1/book.txt
      - From /home/alumno1: dir1/book.txt
      - From /home/alumno1/dir1: book.txt

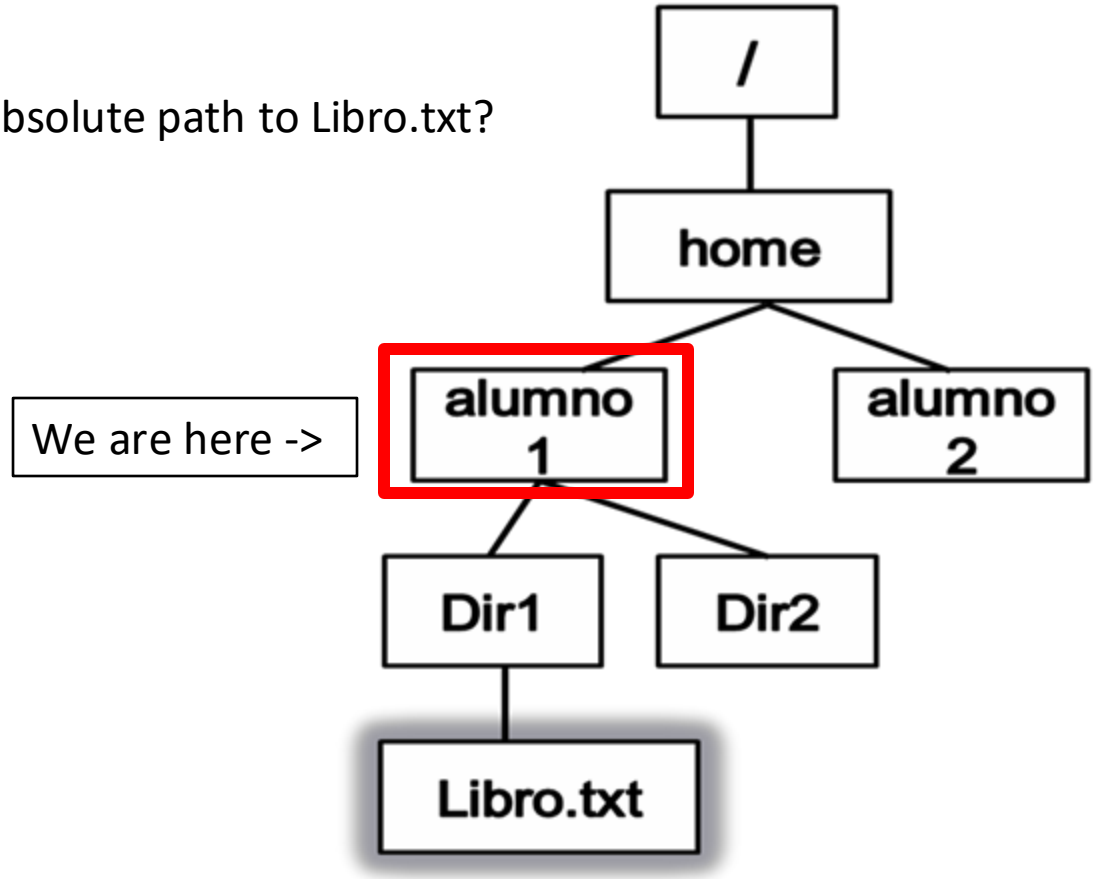
2.4. Linux File System Paths

- Example
  - `../../` To go *home* from *Dir1* or *Dir2*
  - **`../../alumno2`** To go *alumno2* from *Dir1* o *Dir2*.
  - **`../Dir2`** To go from *Dir1* to *Dir2*.



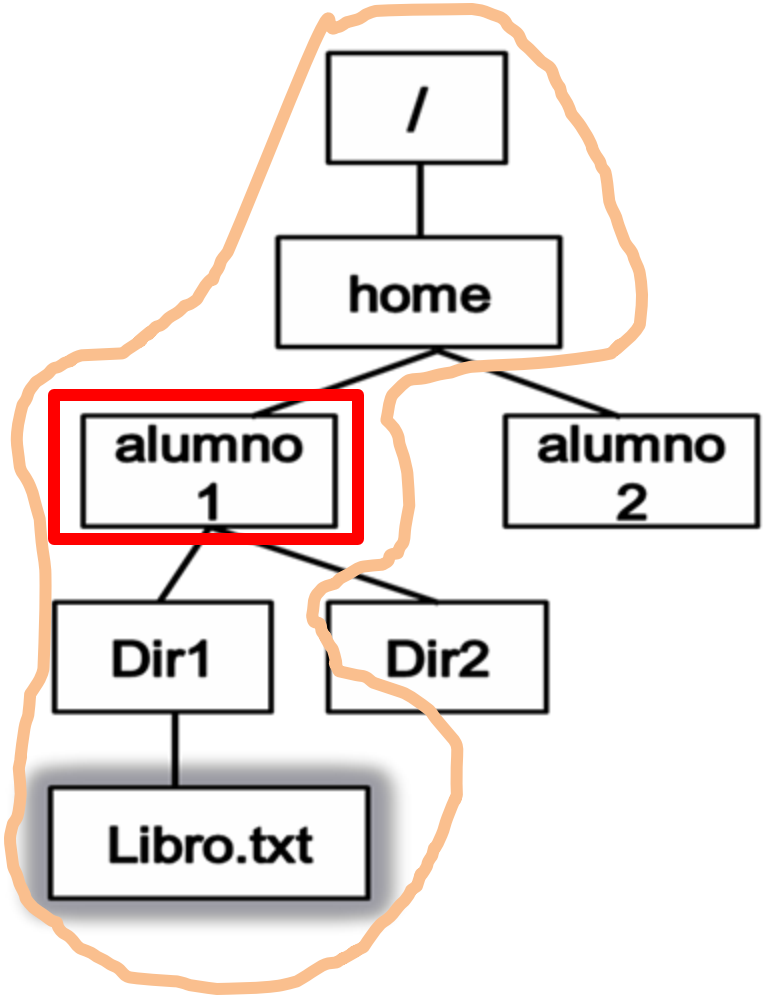
2.4. Linux File System Paths

Which is the absolute path to Libro.txt?



2.4. Linux File System Paths

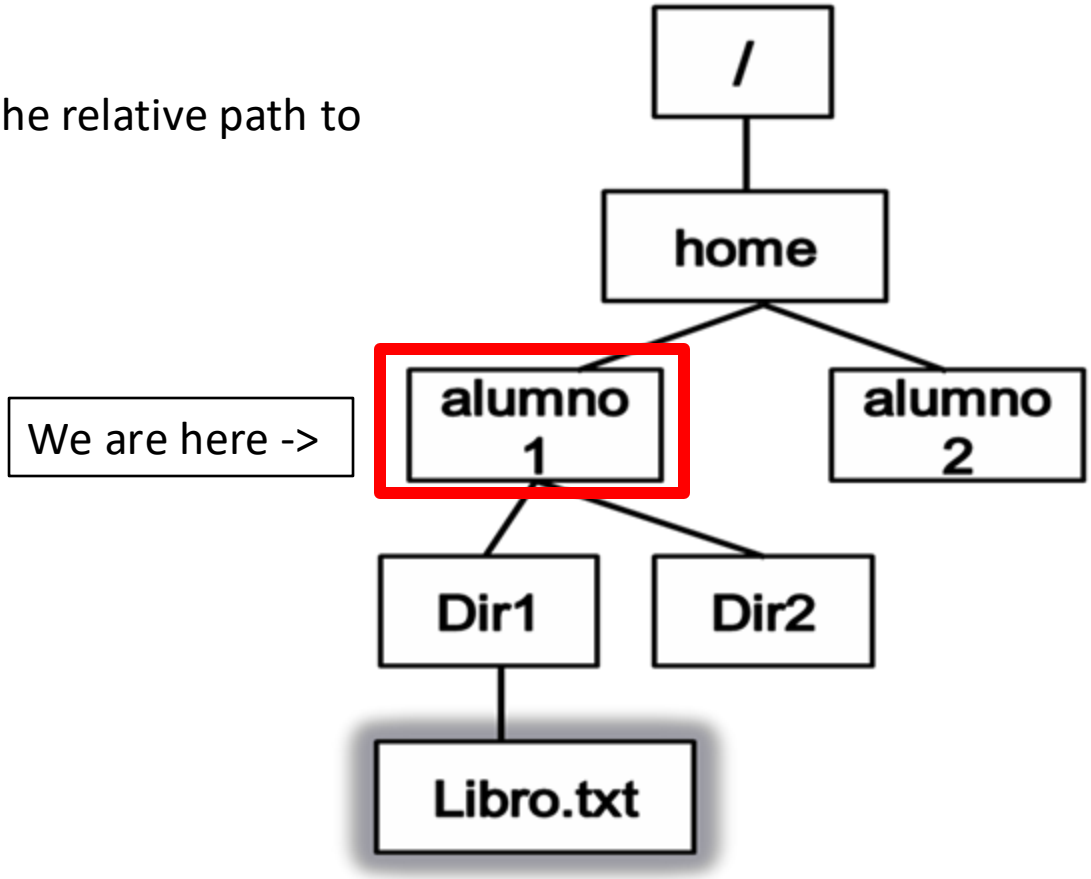
No matter were we are



Absolute path
/home/alumno1/Dir1/Libro.txt

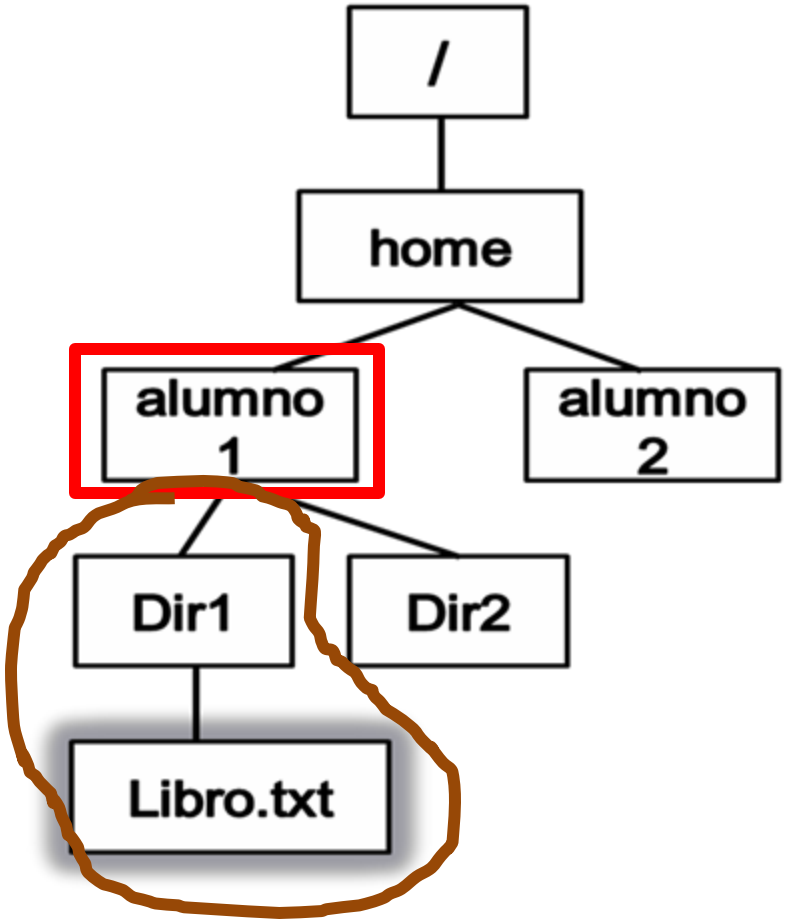
2.4. Linux File System Paths

Which is the the relative path to Libro.txt? ?



2.4. Linux File System Paths

Important pwd or “.”



Relative paths:
./Dir1/Libro.txt
or Dir1/Libro.txt

Overview

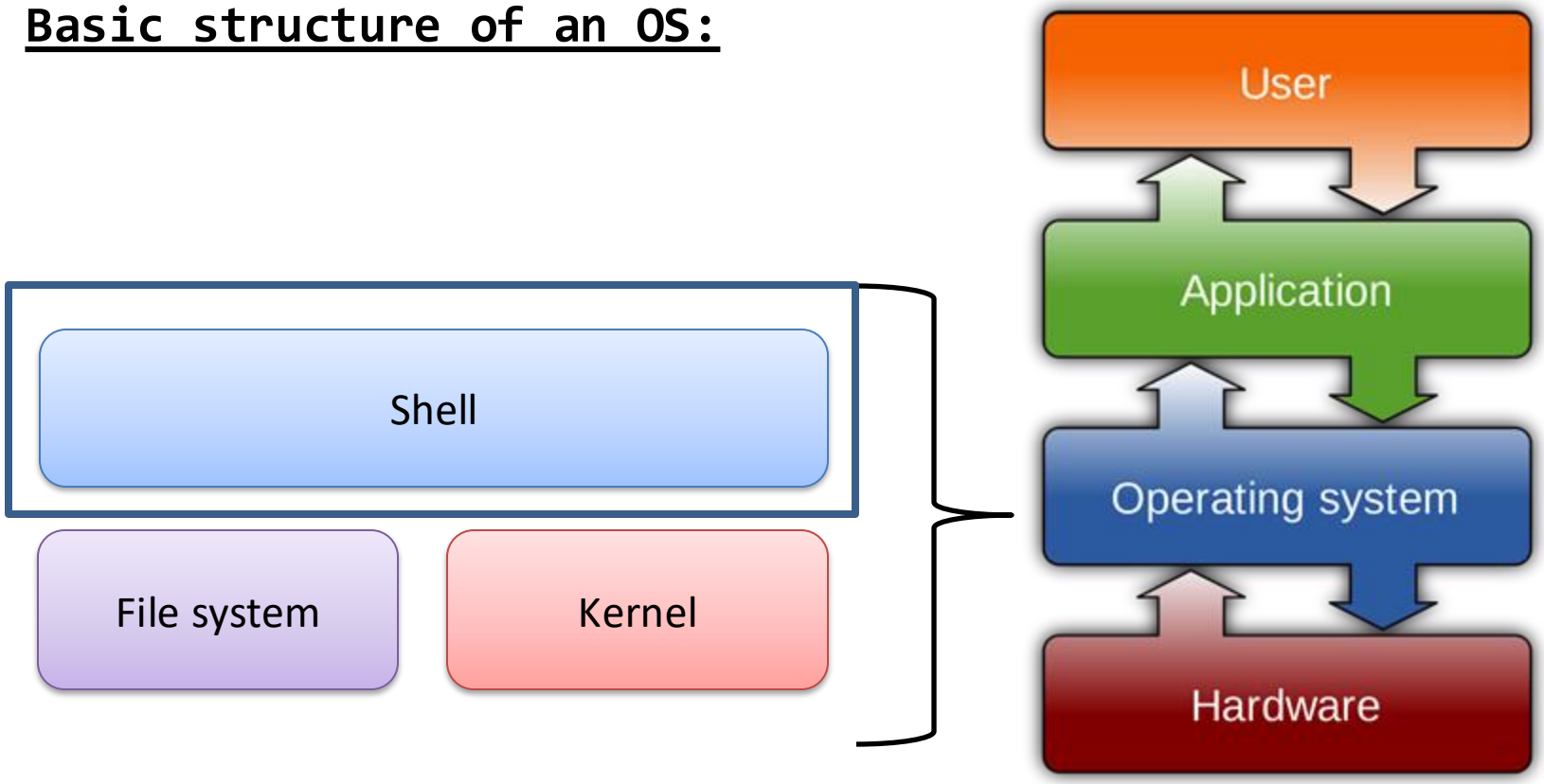
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3. Basic Commands

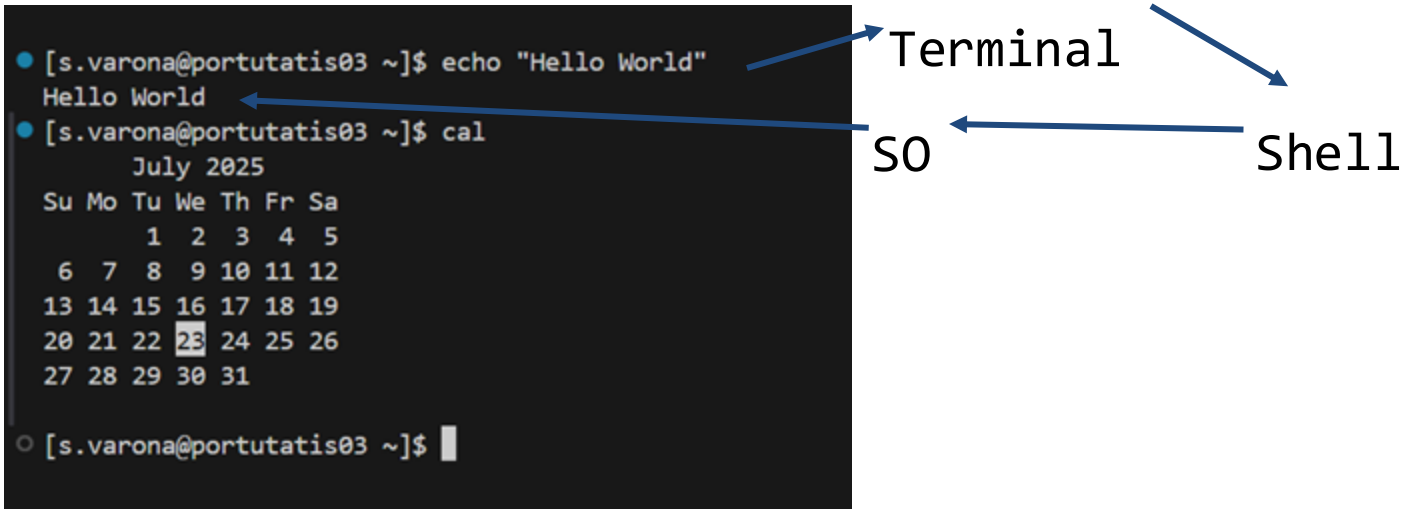
1. Shell
2. Prompt
3. Basic Commands

3.1. Shell

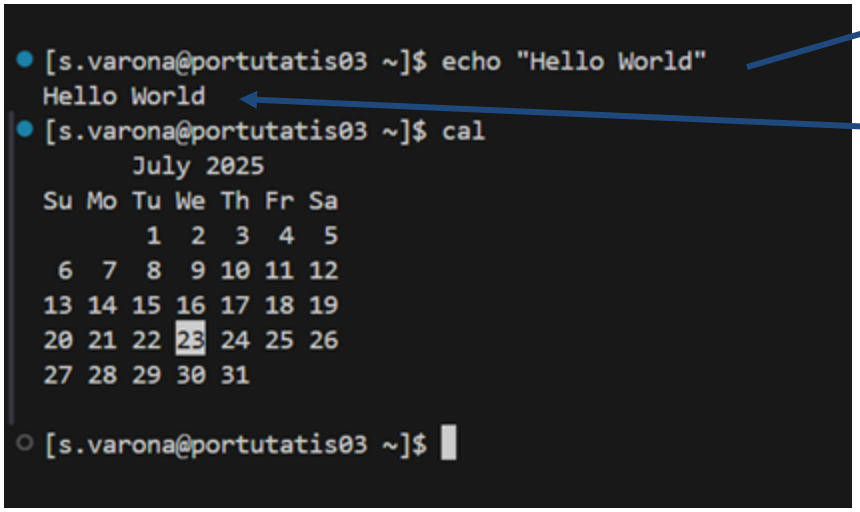
Basic structure of an OS:

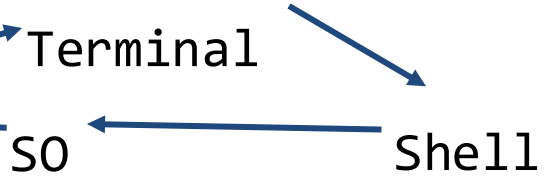


3.1. Shell


[s.varona@portutatis03 ~]$ echo "Hello World"
Hello World
[s.varona@portutatis03 ~]$ cal
July 2025
Su Mo Tu We Th Fr Sa
1 2 3 4 5
6 7 8 9 10 11 12
13 14 15 16 17 18 19
20 21 22 23 24 25 26
27 28 29 30 31
[s.varona@portutatis03 ~]$
Terminal
SO
Shell

3.1. Shell





Bash

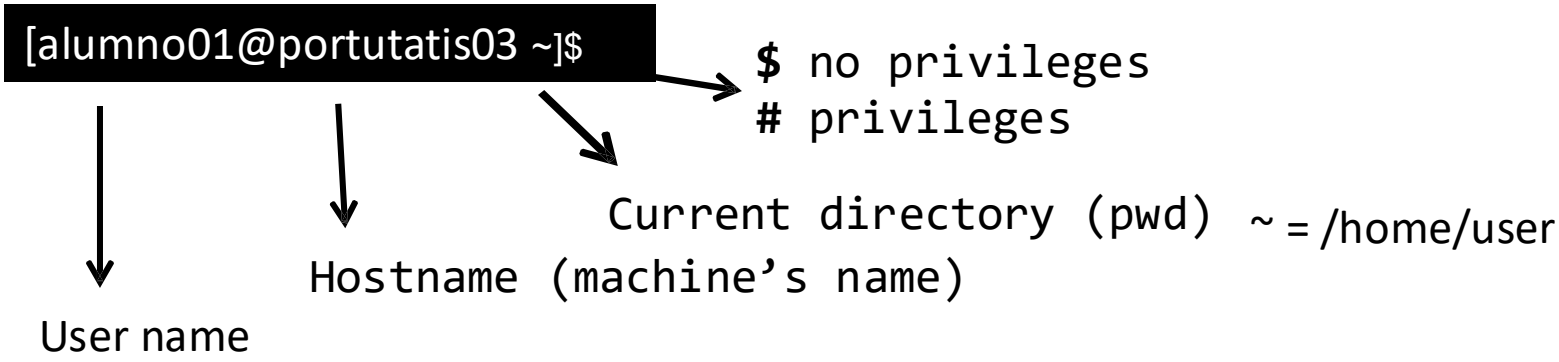
- Navigate the file system
- Run programs and scripts
- Manage processes
- Administer users and permissions
- Automate repetitive tasks

3. Basic Commands

1. Shell
2. Prompt
3. Basic Commands

3.2. Shell's prompt

- Shell waits for the user to write commands in a line called prompt.
- Prompt line gives some important information that can be easily understood:



- This example prompt gives the information that the user is user, which has no admin privileges, which is connected to portutatis03 node in the HPC, and the directory where user is located is the /home/user folder.

3. Basic Commands

1. Shell
2. Prompt
3. Basic Commands

pwd (Print Working Directory)

Shows the current directory

```
[s.varona@portutatis03 ~]$ pwd
/home/s.varona
[s.varona@portutatis03 ~]$
```



ls (List)

List the content of a directory

```
[s.varona@portutatis03 ~]$ pwd
/home/s.varona
[s.varona@portutatis03 ~]$ ls
Descargas  Documentos  imagen.png  script.sh
[s.varona@portutatis03 ~]$
```



cd (Change Directory)

Move from one directory to another

```
[s.varona@portutatis03 ~]$ pwd
/home/s.varona
[s.varona@portutatis03 ~]$ ls
Descargas  Documentos  imagen.png  script
[s.varona@portutatis03 ~]$ cd Documentos/
[s.varona@portutatis03 Documentos]$ pwd
/home/s.varona/Documentos
[s.varona@portutatis03 Documentos]$
```



cd (Change Direcotry)

Move from one directory to another

```
[s.varona@portutatis03 ~]$ pwd
/home/s.varona
[s.varona@portutatis03 ~]$ cd Documentos/
[s.varona@portutatis03 Documentos]$ pwd
/home/s.varona/Documentos
[s.varona@portutatis03 Documentos]$ cd ..
[s.varona@portutatis03 ~]$ pwd
/home/s.varona
[s.varona@portutatis03 ~]$
```

. = Current pwd

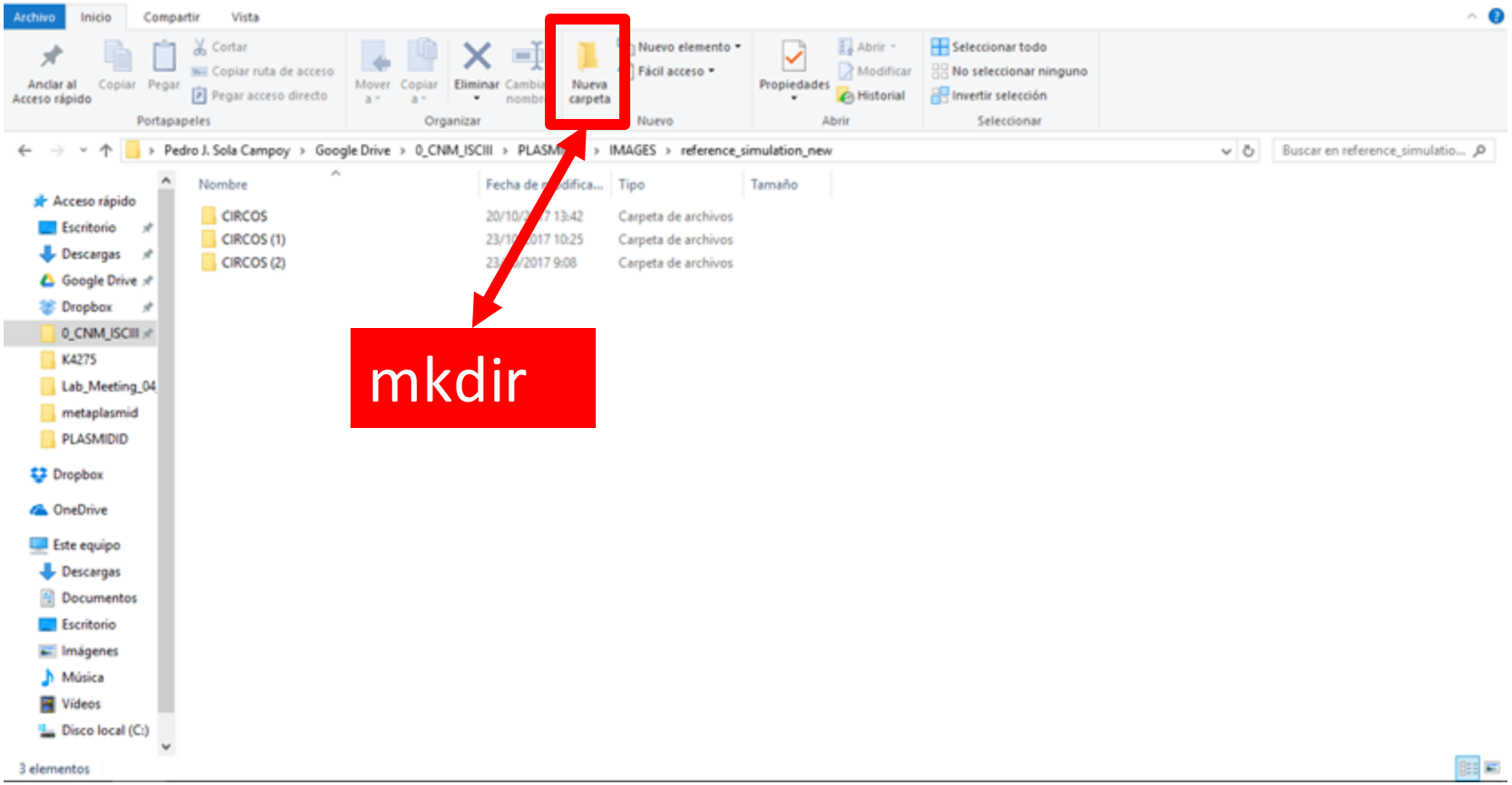
.. = Parent directory

mkdir (Make Directory)

Create an empty directory

```
[s.varona@portutatis03 ~]$ pwd
/home/s.varona
[s.varona@portutatis03 ~]$ ls
Descargas  Documentos  imagen.png  script.sh
[s.varona@portutatis03 ~]$ mkdir Carpeta_prueba
[s.varona@portutatis03 ~]$ ls
Carpeta_prueba  Descargas  Documentos  imagen.png  script.sh
[s.varona@portutatis03 ~]$
```

mkdir (Make Directory)



rm (Remove)

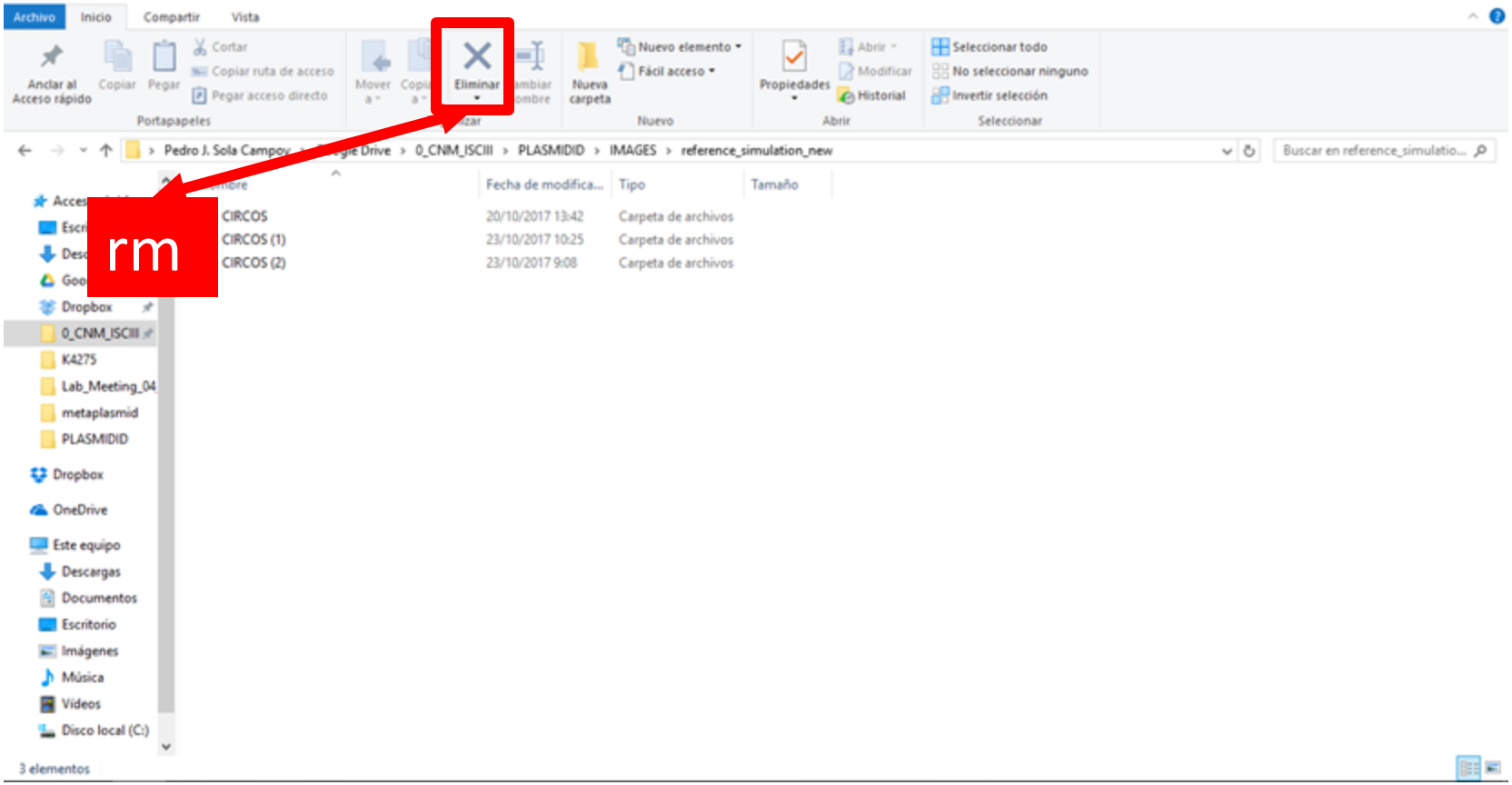
Remove files

```
[s.varona@portutatis03 ~]$ ls
Carpeta_prueba  Descargas  Documentos  imagen.png  script.sh
[s.varona@portutatis03 ~]$ rm imagen.png
[s.varona@portutatis03 ~]$ ls
Carpeta_prueba  Descargas  Documentos  script.sh
[s.varona@portutatis03 ~]$
```



It doesn’t ask for a confirmation

rm (Remove)


Archivo
Inicio
Compartir
Vista
Anclar al Acceso rápido
Copiar
Pegar
Cortar
Copiar ruta de acceso
Pegar acceso directo
Portapapeles
Mover a
Copiar a
Eliminar
Cambiar nombre
Nueva carpeta
Nuevo elemento
Fácil acceso
Nuevo
Propiedades
Abrir
Modificar
Historial
Abrir
Seleccionar todo
No seleccionar ninguno
Invertir selección
Seleccionar
Pedro J. Sola Campoy
Google Drive
0_CNM_ISCIII
PLASMIDID
IMAGES
reference_simulation_new
Buscar en reference_simulatio...
Nombre
Fecha de modifica...
Tipo
Tamaño
CIRCOS
20/10/2017 13:42
Carpeta de archivos
CIRCOS (1)
23/10/2017 10:25
Carpeta de archivos
CIRCOS (2)
23/10/2017 9:08
Carpeta de archivos
rm
Dropbox
0_CNM_ISCIII
K4275
Lab_Meeting_04
metaplasmid
PLASMIDID
Dropbox
OneDrive
Este equipo
Descargas
Documentos
Escritorio
Imágenes
Música
Vídeos
Disco local (C:)
3 elementos

rmdir (Remove Directory)

Remove empty directories

```
[s.varona@portutatis03 ~]$ ls
Carpeta_prueba  Descargas  Documentos  script.sh
[s.varona@portutatis03 ~]$ rmdir Carpeta_prueba/
[s.varona@portutatis03 ~]$ ls
Descargas  Documentos  script.sh
[s.varona@portutatis03 ~]$
```



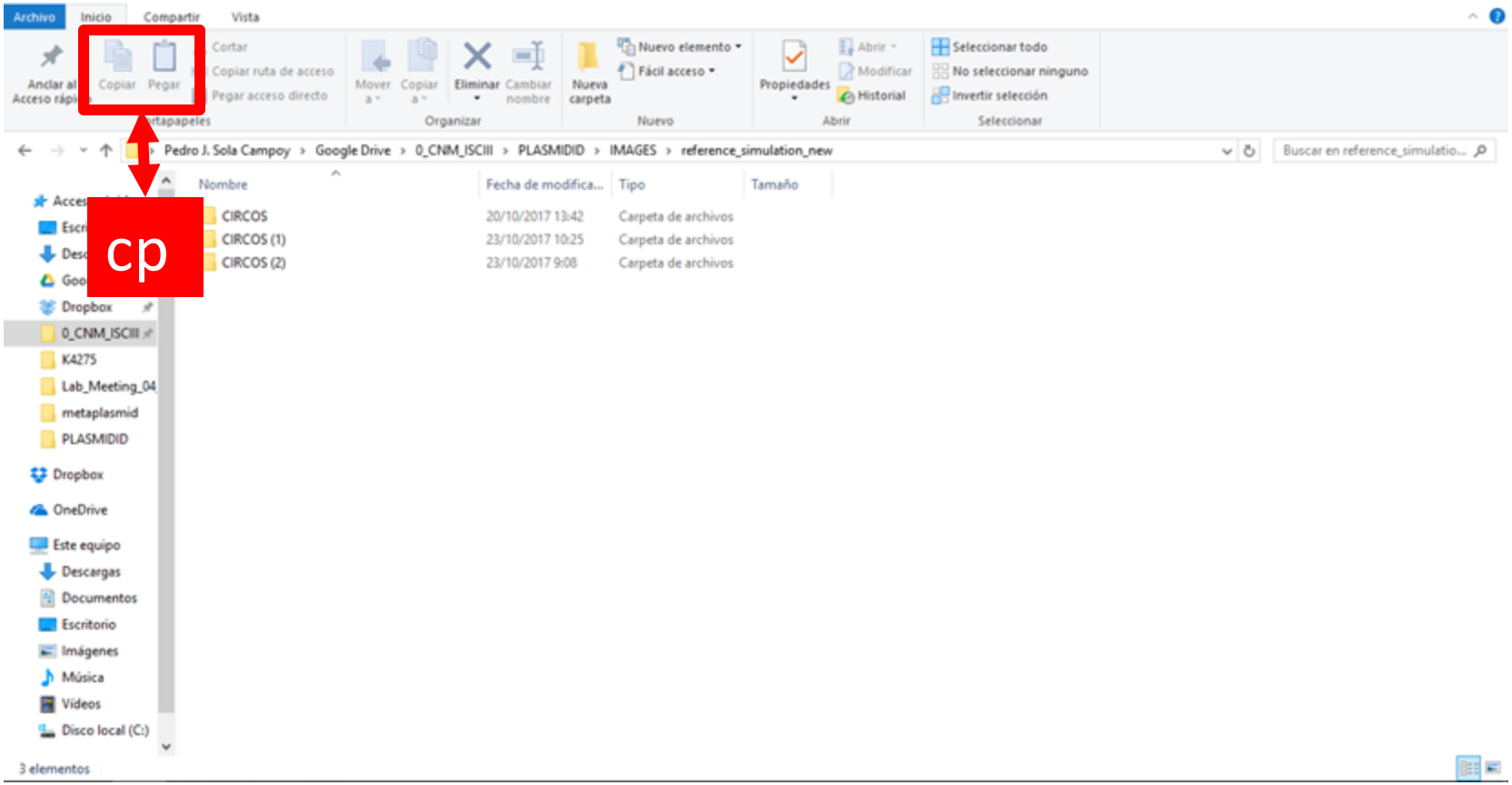
It doesn’t ask for a confirmation

cp (Copy)

Copy a file

```
[s.varona@portutatis03 ~]$ ls
Descargas  Documentos  script.sh
[s.varona@portutatis03 ~]$ cp script.sh copia_script.sh
[s.varona@portutatis03 ~]$ ls
copia_script.sh  Descargas  Documentos  script.sh
[s.varona@portutatis03 ~]$
```

cp (Copy)

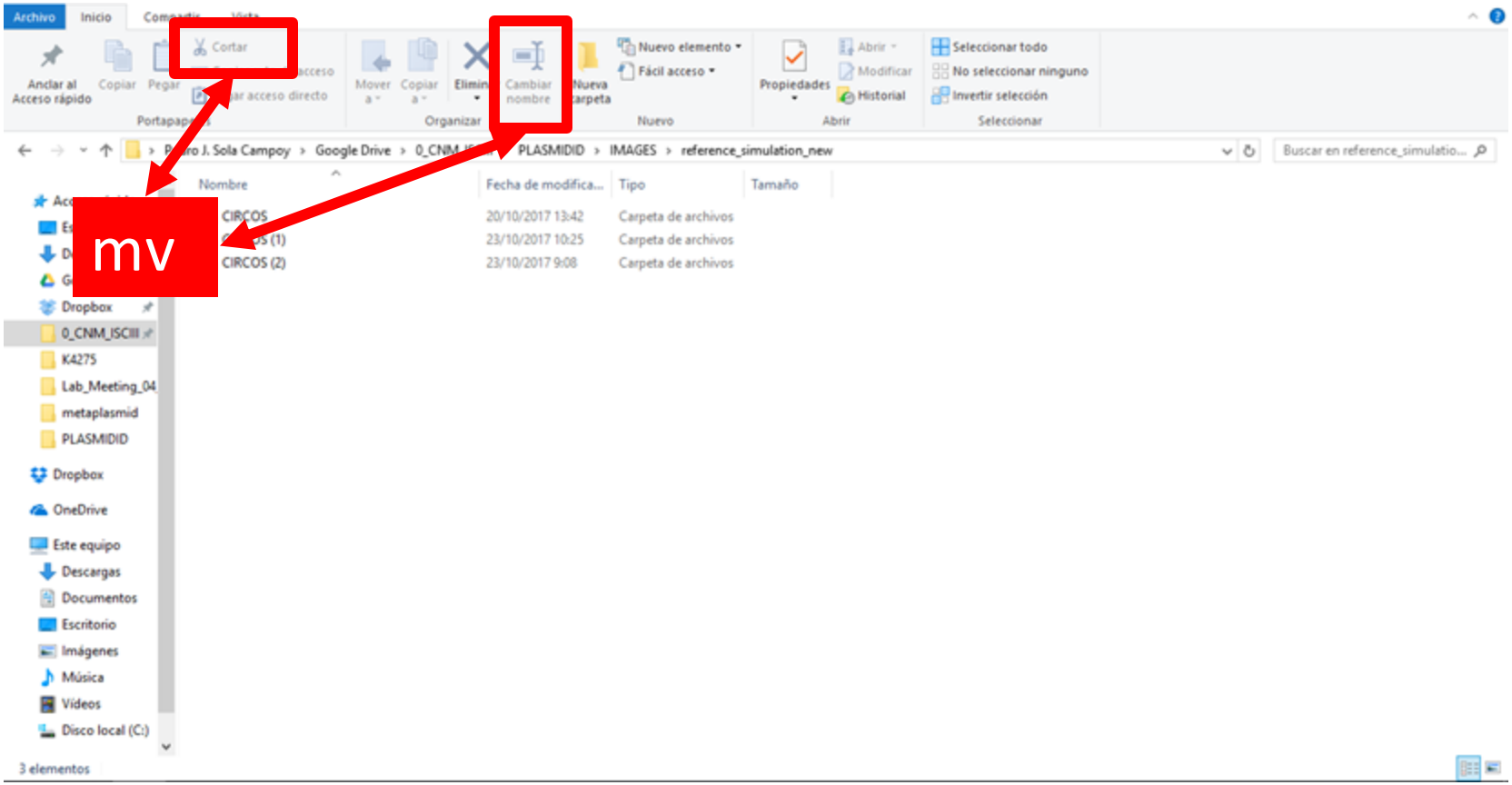

Archivo
Inicio
Compartir
Vista
Anclar al Acceso rápido
Copiar
Pegar
Cortar
Copiar ruta de acceso
Pegar acceso directo
Portapapeles
Mover a
Copiar a
Eliminar
Cambiar nombre
Organizar
Nueva carpeta
Nuevo elemento
Fácil acceso
Nuevo
Propiedades
Abrir
Modificar
Historial
Abrir
Seleccionar todo
No seleccionar ninguno
Invertir selección
Seleccionar
Pedro J. Sola Campoy › Google Drive › 0_CNM_ISCIII › PLASMIDID › IMAGES › reference_simulation_new
Buscar en reference_simulatio...
Nombre
Fecha de modifica...
Tipo
Tamaño
CIRCOS
20/10/2017 13:42
Carpeta de archivos
CIRCOS (1)
23/10/2017 10:25
Carpeta de archivos
CIRCOS (2)
23/10/2017 9:08
Carpeta de archivos
cp
Dropbox
0_CNM_ISCIII
K4275
Lab_Meeting_04
metaplasmid
PLASMIDID
Dropbox
OneDrive
Este equipo
Descargas
Documentos
Escritorio
Imágenes
Música
Vídeos
Disco local (C:)
3 elementos

mv (Move)

Rename a file or move it to another directory

```
[s.varona@portutatis03 ~]$ ls
copia_script.sh  Descargas  Documentos  script.sh
[s.varona@portutatis03 ~]$ mv copia_script.sh script_de_pruebas.sh
[s.varona@portutatis03 ~]$ ls
Descargas  Documentos  script_de_pruebas.sh  script.sh
[s.varona@portutatis03 ~]$ mv script_de_pruebas.sh Documentos/
[s.varona@portutatis03 ~]$ ls
Descargas  Documentos  script.sh
[s.varona@portutatis03 ~]$ ls Documentos/
script_de_pruebas.sh
[s.varona@portutatis03 ~]$
```

mv (Move)

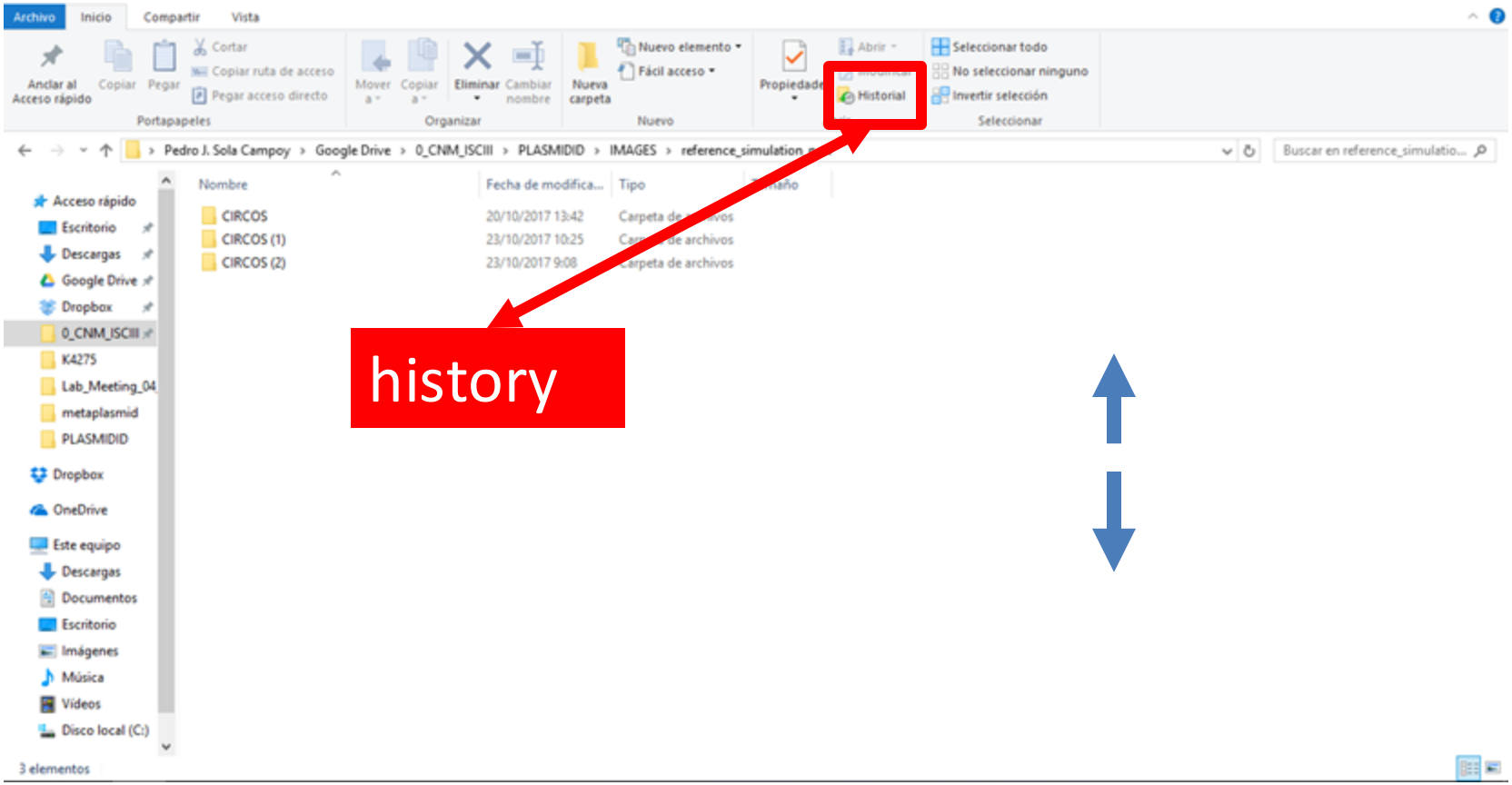

Archivo
Inicio
Cortar
Anclar al Acceso rápido
Copiar
Pegar
Mover a
Copiar a
Cambiar nombre
Nueva carpeta
Nuevo elemento
Fácil acceso
Propiedades
Abrir
Modificar
Historial
Seleccionar todo
No seleccionar ninguno
Invertir selección
Organizar
Nuevo
Abrir
Seleccionar
PLASMIDID
IMAGES
reference_simulation_new
Nombre
Fecha de modifica...
Tipo
Tamaño
CIRCOS
20/10/2017 13:42
Carpeta de archivos
CIRCOS (2)
23/10/2017 9:08
mv
Dropbox
0_CNM_ISCIII
K4275
Lab_Meeting_04
metaplasmid
PLASMIDID
OneDrive
Este equipo
Descargas
Documentos
Escritorio
Imágenes
Música
Vídeos
Disco local (C:)
3 elementos

history

Show history of commands used

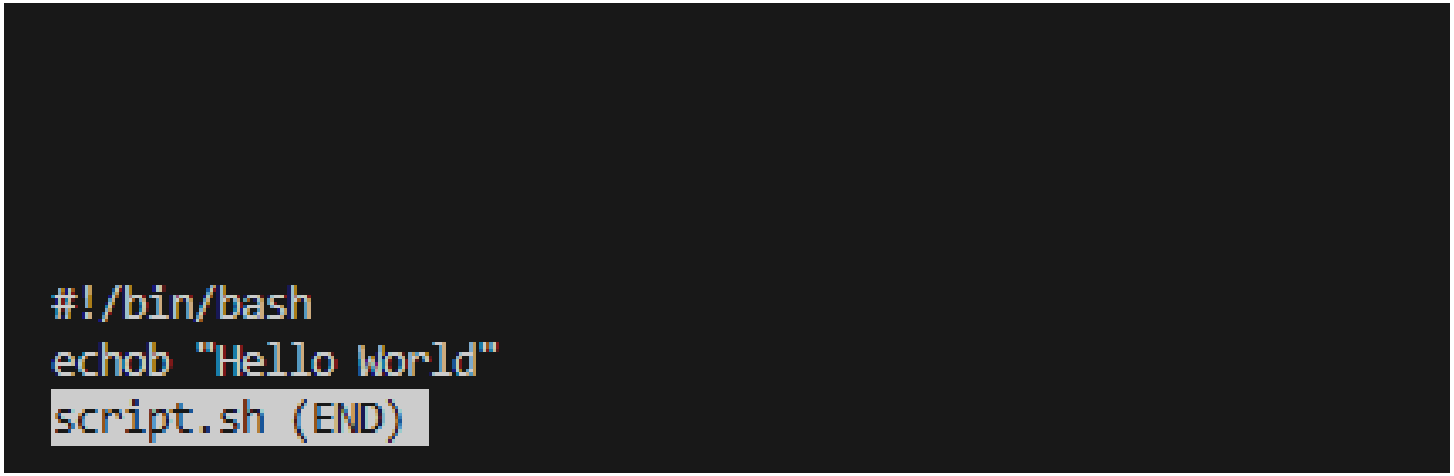
```
[s.varona@portutatis03 ~]$ history
1008  24/07/25 19:58:09 ls
1009  24/07/25 19:58:09 mv copia_script.sh script_de_pruebas.sh
1010  24/07/25 19:58:11 ls
1011  24/07/25 19:58:18 mv script_de_pruebas.sh Documentos/
1012  24/07/25 19:58:20 ls
1013  24/07/25 19:58:22 ls Documentos/
1014  24/07/25 19:58:36 mv Documentos/* Descargas/
1015  24/07/25 19:58:43 mv Descargas/script_de_pruebas.sh Documentos/
1016  24/07/25 19:58:46 ls Documentos/
1017  24/07/25 19:59:48 history
```

history


Archivo
Inicio
Compartir
Vista
Historial
Pedro J. Sola Campoy › Google Drive › 0_CNM_ISCIII › PLASMIDID › IMAGES › reference_simulation_e...
Nombre
Fecha de modifica...
Tipo
CIRCOS
CIRCOS (1)
CIRCOS (2)
20/10/2017 13:42
23/10/2017 10:25
23/10/2017 9:08
Carpeta de archivos
Acceso rápido
Escritorio
Descargas
Google Drive
Dropbox
0_CNM_ISCIII
K4275
Lab_Meeting_04
metaplasmid
PLASMIDID
Dropbox
OneDrive
Este equipo
Descargas
Documentos
Escritorio
Imágenes
Música
Vídeos
Disco local (C:)
3 elementos
history

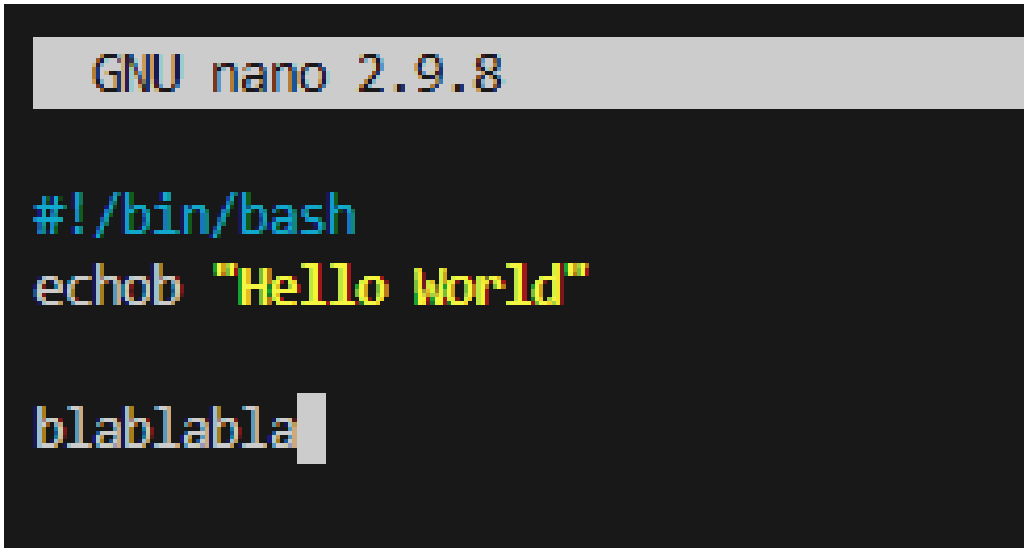
less (Historial)

Show content of a file



nano

Text editor



Save: ctrl+o > Intro

Exit: ctrl+x

Basic Commands

REMEMBER:

TAB is your friend!

Hit it to autocomplete a command, file, path or get suggestion to do it

Overview

1. Linux OS
2. Linux file system
3. Basic commands
4. Command line syntax
5. Linux users and privileges

4. Command Line Syntax

1. Syntax
2. Input
3. Output
4. Redirect
5. Pipes
6. WildCards and Regular expressions

4.1. Command Line Syntax

Linux command line follows a simple syntax common to every command and program you can execute on it:

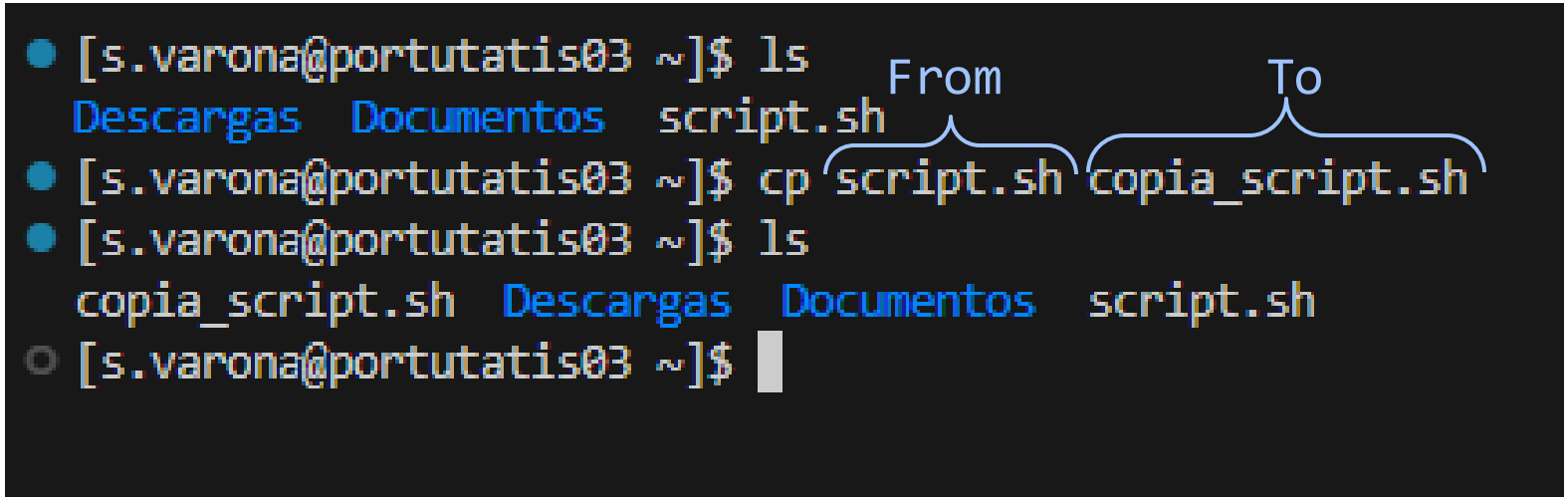
Command [options o parameters] [arguments]

- Options are characters or words preceded by a dash (`ls -la`). They change the way a program works by default.
- Arguments are other parameters that the program may need to run. The most common ones usually are the input files.
- REMEMBER: most programs have a –h or --help option which displays a short description and usage guide.

4.1. Command Line Syntax

```
[s.varona@portutatis03 ~]$ ls
Descargas  Documentos  script.sh
[s.varona@portutatis03 ~]$ ls -l
total 8
drwxrwxr-x 4 s.varona s.varona 4096 Jul 24 19:58 Descargas
drwxrwxr-x 2 s.varona s.varona 4096 Jul 24 19:58 Documentos
-rw-rw-r-- 1 s.varona s.varona   32 Jul 24 19:27 script.sh
[s.varona@portutatis03 ~]$ ls -l -a
total 300
drwx------ 19 s.varona s.varona  4096 Jul 24 19:58 .
drwxr-xr-x 74 root     root      8192 Jul  7 13:11 ..
-rw-------  1 s.varona s.varona 35351 Jul 24 16:09 .bash_history
-rw-r--r--  1 s.varona s.varona    18 Jan 17  2022 .bash_logout
-rw-r--r--  1 s.varona s.varona   141 Jan 17  2022 .bash_profile
drwxrwxr-x  4 s.varona s.varona  4096 Jul 24 19:58 Descargas
drwxrwxr-x  2 s.varona s.varona  4096 Jul 24 19:58 Documentos
-rw-rw-r--  1 s.varona s.varona    32 Jul 24 19:27 script.sh
```

4.1. Command Line Syntax

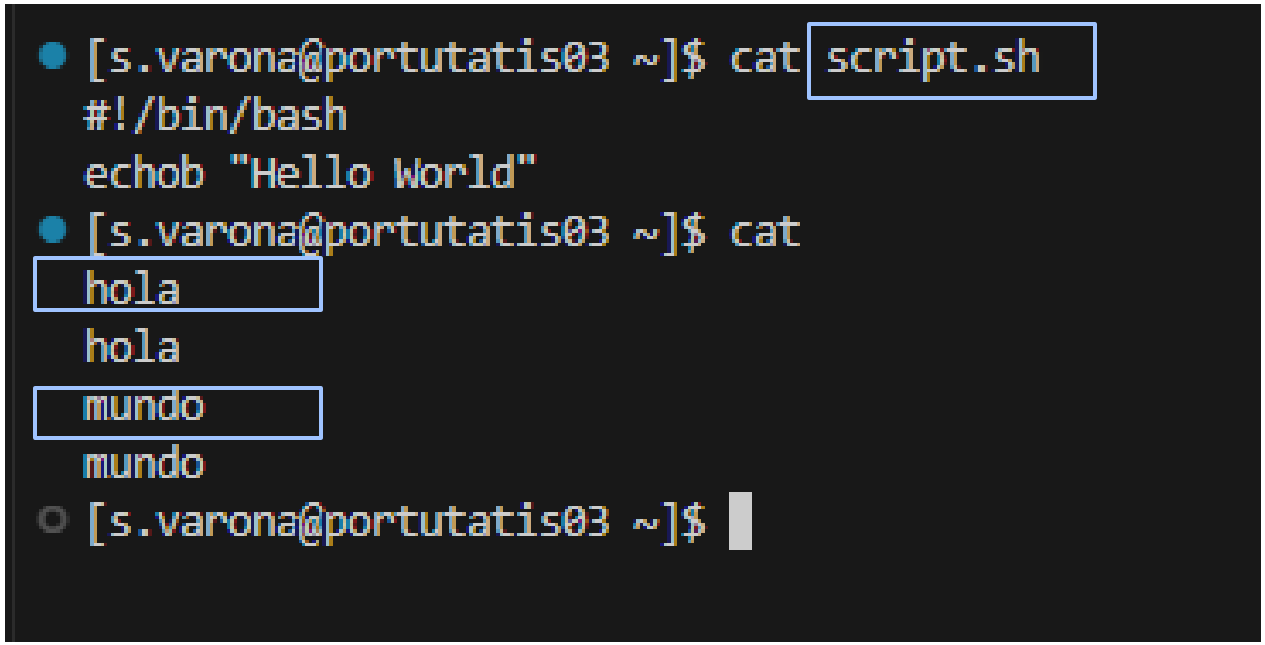


4. Command Line Syntax

1. Syntax
2. Input
3. Output
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4.2. Input

Standard Input (stdin): The data a command receives.



4. Command Line Syntax

1. Syntax
2. Input
3. Output
4. Redirect
5. Pipes
6. WildCards and Regular expressions

4.3. Output

Standard Output (stdout): data a command sends back as a result.

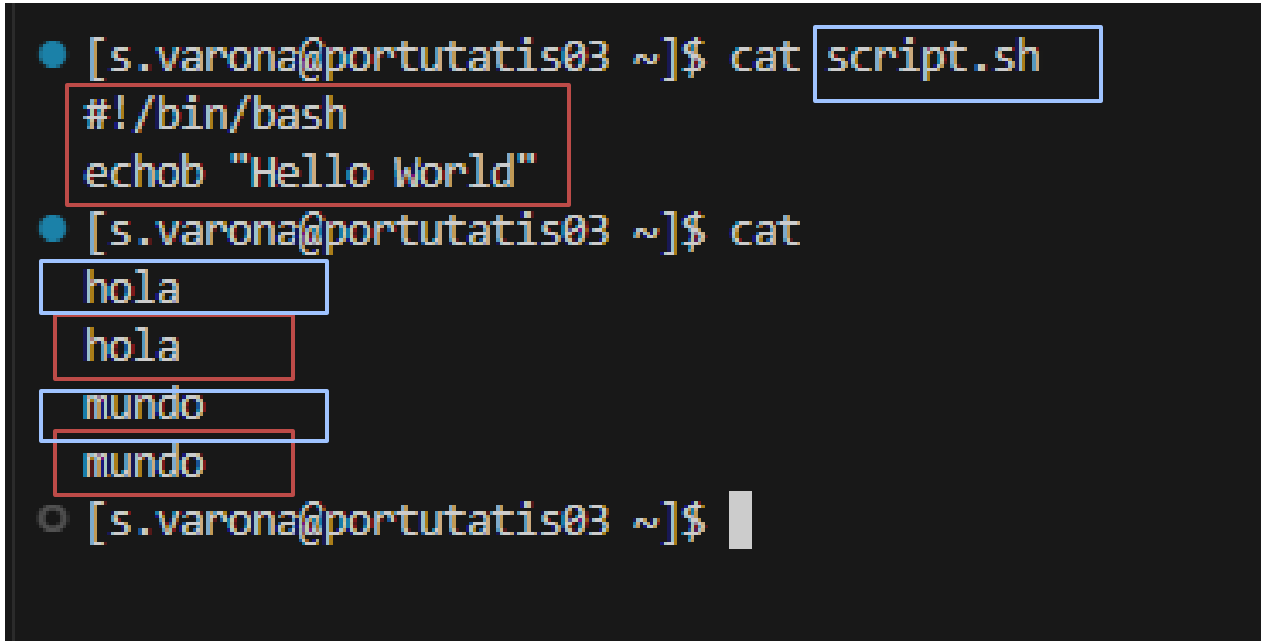
By default, it is printed to the screen.

```
[s.varona@portutatis03 ~]$ cat script.sh
#!/bin/bash
echob "Hello World"
[s.varona@portutatis03 ~]$ cat
hola
hola
mundo
mundo
[s.varona@portutatis03 ~]$
```

Input/Output

stdin

stdout



4. Command Line Syntax

1. Syntax
2. Input
3. Output
4. Redirect
5. Pipes
6. WildCards and Regular expressions

4.4. Redirect

Redirect stdout (>)

```
[s.varona@portutatis03 ~]$ echo "Hola mundo" > saludo.txt
[s.varona@portutatis03 ~]$ cat saludo.txt
Hola mundo
[s.varona@portutatis03 ~]$
```

4.4. Redirect

Redirect stdout at the end (>>)

```
[s.varona@portutatis03 ~]$ echo "Hola mundo" > saludo.txt
[s.varona@portutatis03 ~]$ cat saludo.txt
Hola mundo
[s.varona@portutatis03 ~]$ echo "Otra línea" >> saludo.txt
[s.varona@portutatis03 ~]$ cat saludo.txt
Hola mundo
Otra línea
[s.varona@portutatis03 ~]$
```

4.4. Redirect

Input from file (<)

```
[s.varona@portutatis03 ~]$ cat < saludo.txt
Hola mundo
Otra línea
[s.varona@portutatis03 ~]$
```

4.4. Redirect

Redirect error log (2>)

```
[s.varona@portutatis03 ~]$ cat archivo_que_no_existe.txt 2> error.txt
[s.varona@portutatis03 ~]$ cat error.txt
cat: archivo_que_no_existe.txt: No such file or directory
[s.varona@portutatis03 ~]$ cat archivo_que_no_existe.txt
cat: archivo_que_no_existe.txt: No such file or directory
[s.varona@portutatis03 ~]$
```

4.4. Redirect

Redirect error log and output (2>&1)

```
[s.varona@portutatis03 ~]$ cat archivo_que_no_existe.txt > salida.txt 2>&1
[s.varona@portutatis03 ~]$ cat salida.txt
cat: archivo_que_no_existe.txt: No such file or directory
[s.varona@portutatis03 ~]$
```

4. Command Line Syntax

1. Syntax
2. Input
3. Output
4. Redirect
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4.5. Pipes

Connect stdout from a command to stdin of another command

```
[Command A] --output--> | --input--> [Command B]
```

4.5. Pipes

Connect stdout from a command to stdin of another command

```
[Command A] --output--> | --input--> [Command B]
```

```
[s.varona@portutatis03 ~]$ ls -r
textos.txt  script.sh  saludo.txt  salida.txt  error.txt  Documentos  Descargas
[s.varona@portutatis03 ~]$ ls -r | sort
Descargas
Documentos
error.txt
salida.txt
saludo.txt
script.sh
textos.txt
```

4.5. Pipes

Connect stdout from a command to stdin of another command

```
[Command A] --output--> | --input--> [Command B]
```

```
[s.varona@portutatis03 ~]$ ls -r
textos.txt  script.sh  saludo.txt  salida.txt  error.txt  Documentos  Descargas
[s.varona@portutatis03 ~]$ ls -r | sort
Descargas
Documentos
error.txt
salida.txt
saludo.txt
script.sh
textos.txt
[s.varona@portutatis03 ~]$ ls | wc -l
7
[s.varona@portutatis03 ~]$
```

4.5. Pipes

```
[s.varona@portutatis03 ~]$ ls
Descargas  Documentos  error.txt  salida.txt  saludo.txt  script.sh
[s.varona@portutatis03 ~]$ cat *.txt *.sh  | sort > textos.txt
[s.varona@portutatis03 ~]$ ls
Descargas  Documentos  error.txt  salida.txt  saludo.txt  script.sh  textos.txt
[s.varona@portutatis03 ~]$ cat textos.txt
#!/bin/bash
cat: archivo_que_no_existe.txt: No such file or directory
cat: archivo_que_no_existe.txt: No such file or directory
echob "Hello World"
Hola mundo
Otra línea
[s.varona@portutatis03 ~]$
```

4. Command Line Syntax

1. Syntax
2. Input
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4.6. WildCards and Regular expressions

Patterns used to search, match, or validate text in strings

Pattern	Meaning	Use example	What finds
*	Zero or more repetitions	file*.txt	fileee.txt file1.txt files.txt
^	Start of the string	^file	file.pdf file.txt files.docx
$	End of the string	.txt$	file.txt document.txt
\t	Tabular	Name\tAge John\t25	Name Age John 25
\s	White space	John\sSmith	John Smith

Overview

1. Linux OS
2. Linux file system
3. Basic commands
4. Command line syntax
5. Linux users and privileges

5. Linux users and privileges

1. Users
2. Permissions
3. Change permissions

5.1. Linux Users

- Access entity to system resources
- Each user has:
  - A username
  - A user identifier (UID)
  - A primary group (GID)
  - A home directory (for example, /home/user)
  - A command-line interpreter (shell, such as /bin/bash)
- Type of entities:
  - Normal user:
    - Limited access
    - Only modify owner files
  - Root user:
    - Sysadmin
    - Total access
    - Un/Install software
  - Groups:
    - Group of users to manage collective permissions.

5.1. Linux Users

- Users can be linked to a person or computer process
- Every user may belong to one or more groups
- Every user may has a home folder inside /home
- Users own the files they create, directly or indirectly
- Users can change permissions on files they own
- Users also own processes they execute
- Root rules over them all
- Root home folder is in /root

5. Linux users and privileges

1. Users
2. Permissions
3. Change permissions

5.2. Linux Permissions

To view file permissions and ownership on files and directories, use the **ls -l** command. For example:

```
drwxrwxr-x 2 s.varona s.varona 4096 Jul 24 19:58 Documentos
```

`drwxrwxr-x` are the permissions
`2` is the number of files or directories
`user` is the owner
`user` is the group
`4096` is the size in bytes
`Jan  9 10:11` is the date/time of last access
`documents` is the directory

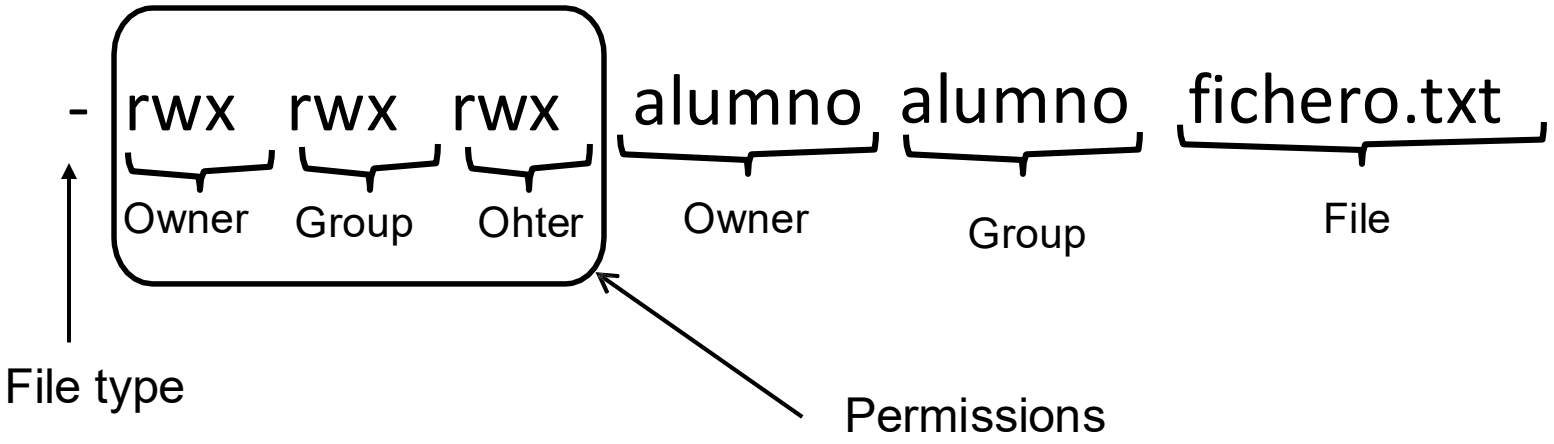
5.2. Linux Permissions

Permissions are the “rights” to act on a file or directory. There are only 3 basic permissions:

- **Read (r)** - allows the contents of the file to be viewed. A read permission on a directory allows you to list the contents of a directory.
- **Write (w)** - allows you to modify the contents of that file. For a directory, the write permission allows you to edit the contents of a directory.
- **Execute (x)** - for a file, the executable permission allows you to run the file and execute a program or script. For a directory, the execute permission allows you to enter the directory and make it your current working directory (pwd or “.”).

5.2. Linux Permissions

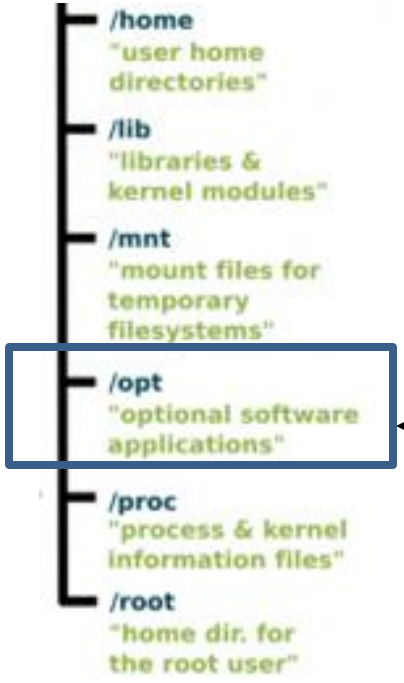
- A file's rights can only be modified by the owner of the file, the group owning the file and the root
- The system stores this right information in a 9 bits sequence.
- This sequence has a sequence of 3 elements for each 3 groups:



5.2. Linux Permissions

- Examples:
  - Example 1: /opt directory

5.2. Linux Permissions



Directorio que contiene software adicional. En nuestro caso aquí se realiza la instalación de software bioinformático

5.2. Linux Permissions

- Examples:
  - Example 1: /opt directory:

```
drwxr-xr-x    root    root    opt
```

  - Owner (root) can read, write and execute
  - Group (root) and rest can only read and execute

5.2. Linux Permissions

- Examples:
  - Example 2: personal directory : /home/alumno

```
drwx------  alumno  clase  alumno
```

  - Owner (alumno) can read, modify and access the directory
  - Group (clase) and rest can’t do anything

5.2. Linux Permissions

- Examples:
  - Example 3: /tmp :

```
drwxrwxrwx    root    root    tmp
```

    - ???

5.2. Linux Permissions

- Examples:
  - Example 3: tmp :

```
drwxrwxrwx    root    root    tmp
```

    - Everybody has permissions for everything

5.2. Linux Permissions

To view file permissions and ownership on files and directories, use the **ls -al** command. For example:

```
drwxr-xr-x 2 user user 4096 Jan  9 10:11 documents
```

`drwxr-xr-x` are the permissions
`2` is the number of files or directories
`user` is the owner
`user` is the group
`4096` is the size in bytes
`Jan  9 10:11` is the date/time of last access
`documents` is the directory

5.2. Linux Permissions

Following previous example:

```
drwxrw-r-- 2 user user 4096 Jan  9 10:11 documents
```

?????

5.2. Linux Permissions

Following previous example:

```
drwxrw-r-- 2 user user 4096 Jan  9 10:11 documents
```

Permissions are listed in the first 10 characters-dash section. The section can be read as follows:

`d` is a directory (`-` for files)
`rwx` the user has read, write, and execute permissions
`rw-` the group has read and write permissions
`r--` all others have read only permissions

5. Linux users and privileges

1. Users
2. Permissions
3. Change permissions

5.3. Modify Permissions

You can only change permission on files you own, while root can change permissions on any file of the system.

To change permissions, use the commands:

- `chmod` – change permissions

```
[s.varona@portutatis03 ~]$ ll
total 12
drwxrwxr-x 4 s.varona s.varona 4096 Jul 24 19:58 Descargas
drwxrwxr-x 2 s.varona s.varona 4096 Jul 24 19:58 Documentos
-rw-rw-r-- 1 s.varona s.varona   58 Jul 24 20:42 error.txt
-rw-rw-r-- 1 s.varona s.varona   58 Jul 24 20:43 salida.txt
-rw-rw-r-- 1 s.varona s.varona   23 Jul 24 20:40 saludo.txt
-rw-rw-r-- 1 s.varona s.varona   32 Jul 24 19:27 script.sh
-rw-rw-r-- 1 s.varona s.varona  171 Jul 24 20:47 textos.txt
[s.varona@portutatis03 ~]$ chmod 755 textos.txt
[s.varona@portutatis03 ~]$ ll
total 12
drwxrwxr-x 4 s.varona s.varona 4096 Jul 24 19:58 Descargas
drwxrwxr-x 2 s.varona s.varona 4096 Jul 24 19:58 Documentos
-rw-rw-r-- 1 s.varona s.varona   58 Jul 24 20:42 error.txt
-rw-rw-r-- 1 s.varona s.varona   58 Jul 24 20:43 salida.txt
-rw-rw-r-- 1 s.varona s.varona   23 Jul 24 20:40 saludo.txt
-rw-rw-r-- 1 s.varona s.varona   32 Jul 24 19:27 script.sh
-rwxr-xr-x 1 s.varona s.varona  171 Jul 24 20:47 textos.txt
[s.varona@portutatis03 ~]$
```

`rwx` to owner (7 = 4 + 2 + 1)

`rx` to group (5 = 4 + 0 + 1)

`rx` to others (5 = 4 + 0 + 1)

5.3. Modify Permissions

- `chown` - change owner

```
[s.varona@portutatis03 ~]$ ll
total 12
drwxrwxr-x 4 s.varona s.varona 4096 Jul 24 19:58 Descargas
drwxrwxr-x 2 s.varona s.varona 4096 Jul 24 19:58 Documentos
-rw-rw-r-- 1 s.varona s.varona   58 Jul 24 20:42 error.txt
-rw-rw-r-- 1 s.varona s.varona   58 Jul 24 20:43 salida.txt
-rw-rw-r-- 1 s.varona s.varona   23 Jul 24 20:40 saludo.txt
-rw-rw-r-- 1 s.varona s.varona   32 Jul 24 19:27 script.sh
-rwxr-xr-x 1 s.varona s.varona  171 Jul 24 20:47 textos.txt
[s.varona@portutatis03 ~]$ chown s.varona:bi textos.txt
[s.varona@portutatis03 ~]$ ll
total 12
drwxrwxr-x 4 s.varona s.varona 4096 Jul 24 19:58 Descargas
drwxrwxr-x 2 s.varona s.varona 4096 Jul 24 19:58 Documentos
-rw-rw-r-- 1 s.varona s.varona   58 Jul 24 20:42 error.txt
-rw-rw-r-- 1 s.varona s.varona   58 Jul 24 20:43 salida.txt
-rw-rw-r-- 1 s.varona s.varona   23 Jul 24 20:40 saludo.txt
-rw-rw-r-- 1 s.varona s.varona   32 Jul 24 19:27 script.sh
-rwxr-xr-x 1 s.varona bi        171 Jul 24 20:47 textos.txt
[s.varona@portutatis03 ~]$
```

5.3. Modify Permissions

- `chown` - change owner

```
[s.varona@portutatis03 ~]$ ll
total 12
drwxrwxr-x 4 s.varona s.varona 4096 Jul 24 19:58 Descargas
drwxrwxr-x 2 s.varona s.varona 4096 Jul 24 19:58 Documentos
-rw-rw-r-- 1 s.varona s.varona   58 Jul 24 20:42 error.txt
-rw-rw-r-- 1 s.varona s.varona   58 Jul 24 20:43 salida.txt
-rw-rw-r-- 1 s.varona s.varona   23 Jul 24 20:40 saludo.txt
-rw-rw-r-- 1 s.varona s.varona   32 Jul 24 19:27 script.sh
-rwxr-xr-x 1 s.varona s.varona  171 Jul 24 20:47 textos.txt
[s.varona@portutatis03 ~]$ chown s.varona:bi textos.txt
[s.varona@portutatis03 ~]$ ll
total 12
drwxrwxr-x 4 s.varona s.varona 4096 Jul 24 19:58 Descargas
drwxrwxr-x 2 s.varona s.varona 4096 Jul 24 19:58 Documentos
-rw-rw-r-- 1 s.varona s.varona   58 Jul 24 20:42 error.txt
-rw-rw-r-- 1 s.varona s.varona   58 Jul 24 20:43 salida.txt
-rw-rw-r-- 1 s.varona s.varona   23 Jul 24 20:40 saludo.txt
-rw-rw-r-- 1 s.varona s.varona   32 Jul 24 19:27 script.sh
-rwxr-xr-x 1 s.varona bi        171 Jul 24 20:47 textos.txt
[s.varona@portutatis03 ~]$
```

5.3. Modify Permissions

Permissions are usually managed in octal format

Every 3 characters group belonging to a set of permissions translates to a number ranging from 0 (---) to 7 (rwx), where

- r=4
- w=2
- x=1

Example:

```
rwx = 4 + 2 + 1 = 7
rw- = 4 + 2 + 0 = 6
r-- = 4 + 0 + 0 = 4
```

Thanks for your attention!

